

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

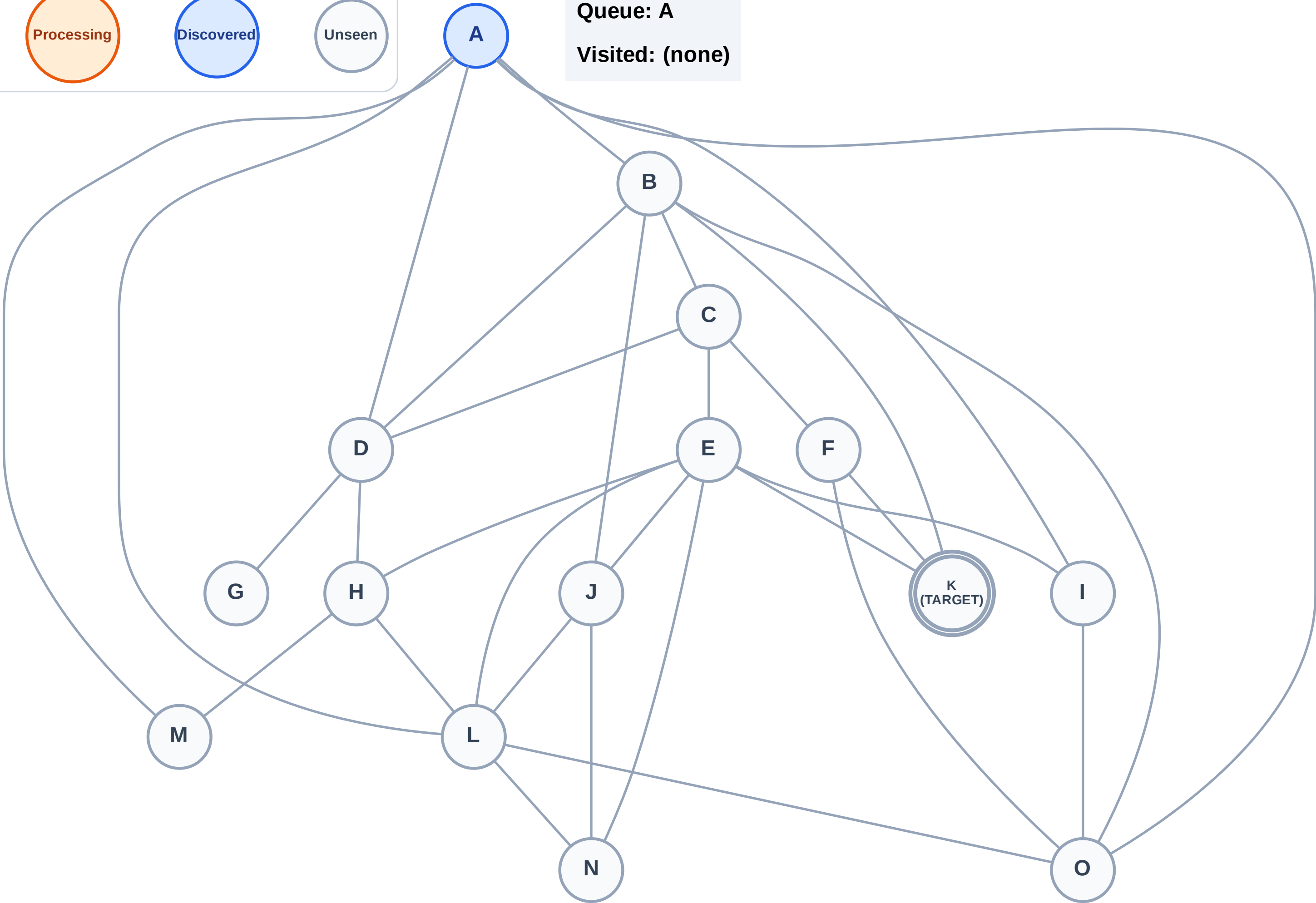
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

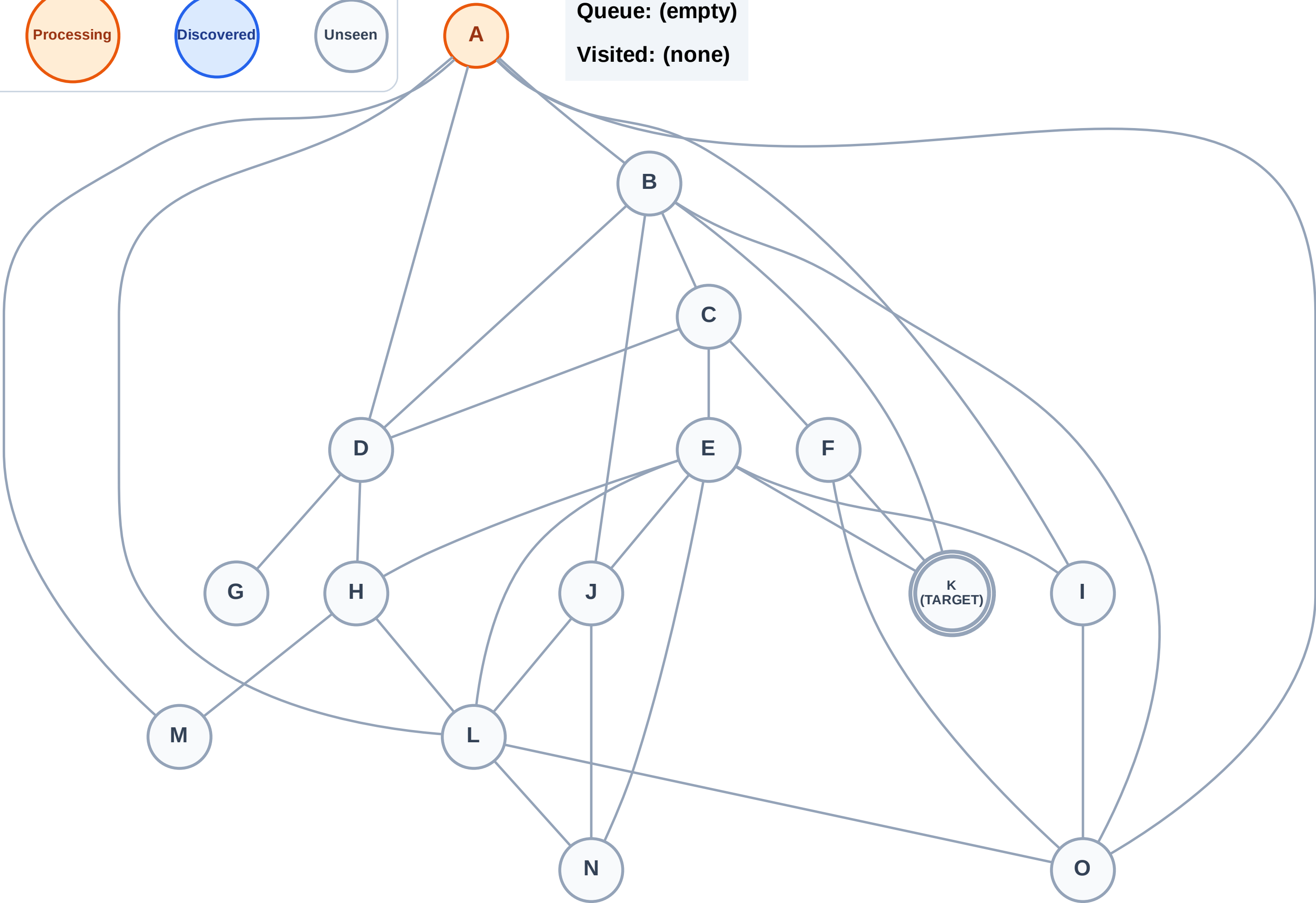
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, D, I, L, M, O from A

Legend

Complete

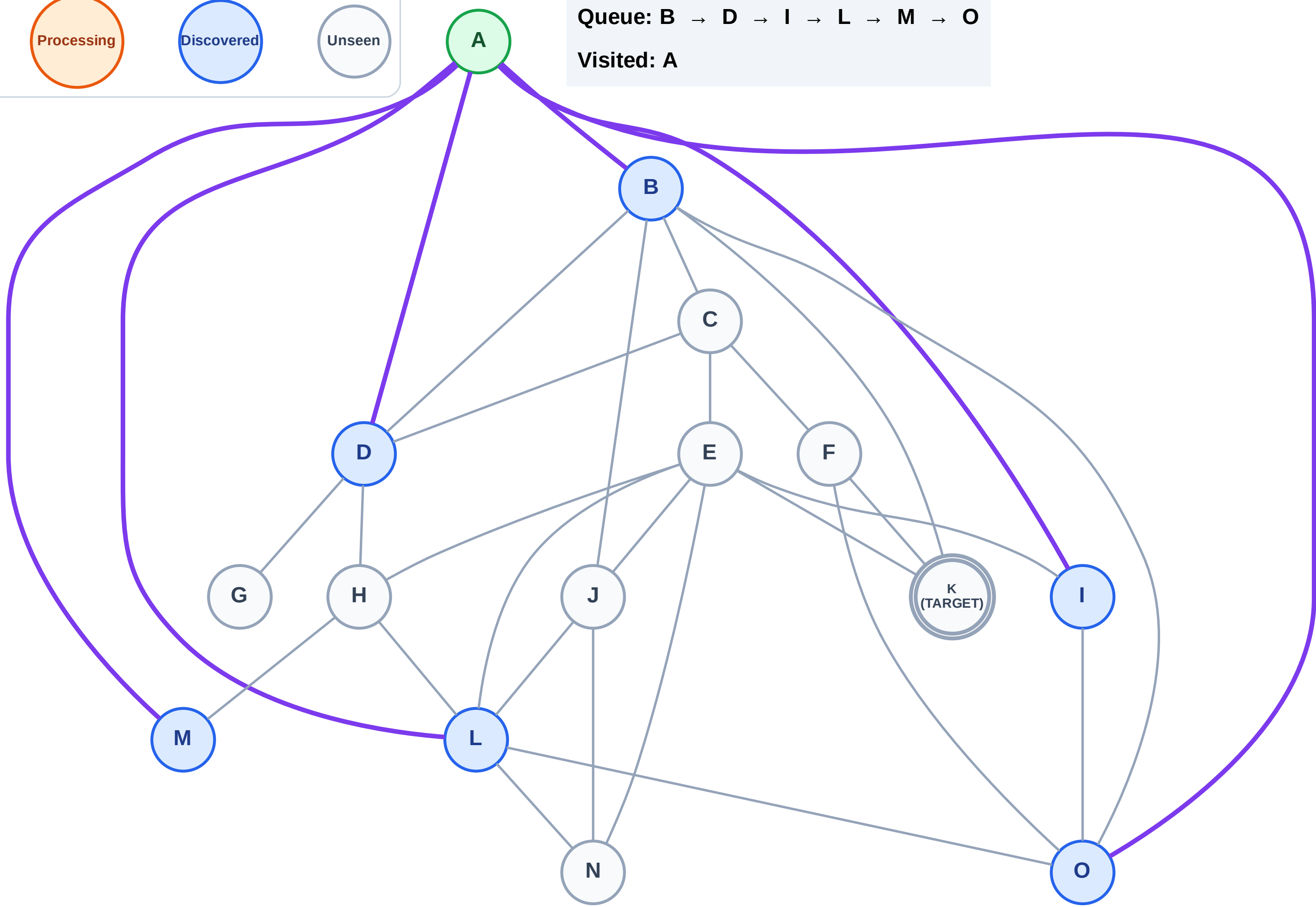
Processing

Discovered

Unseen

Queue: B → D → I → L → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

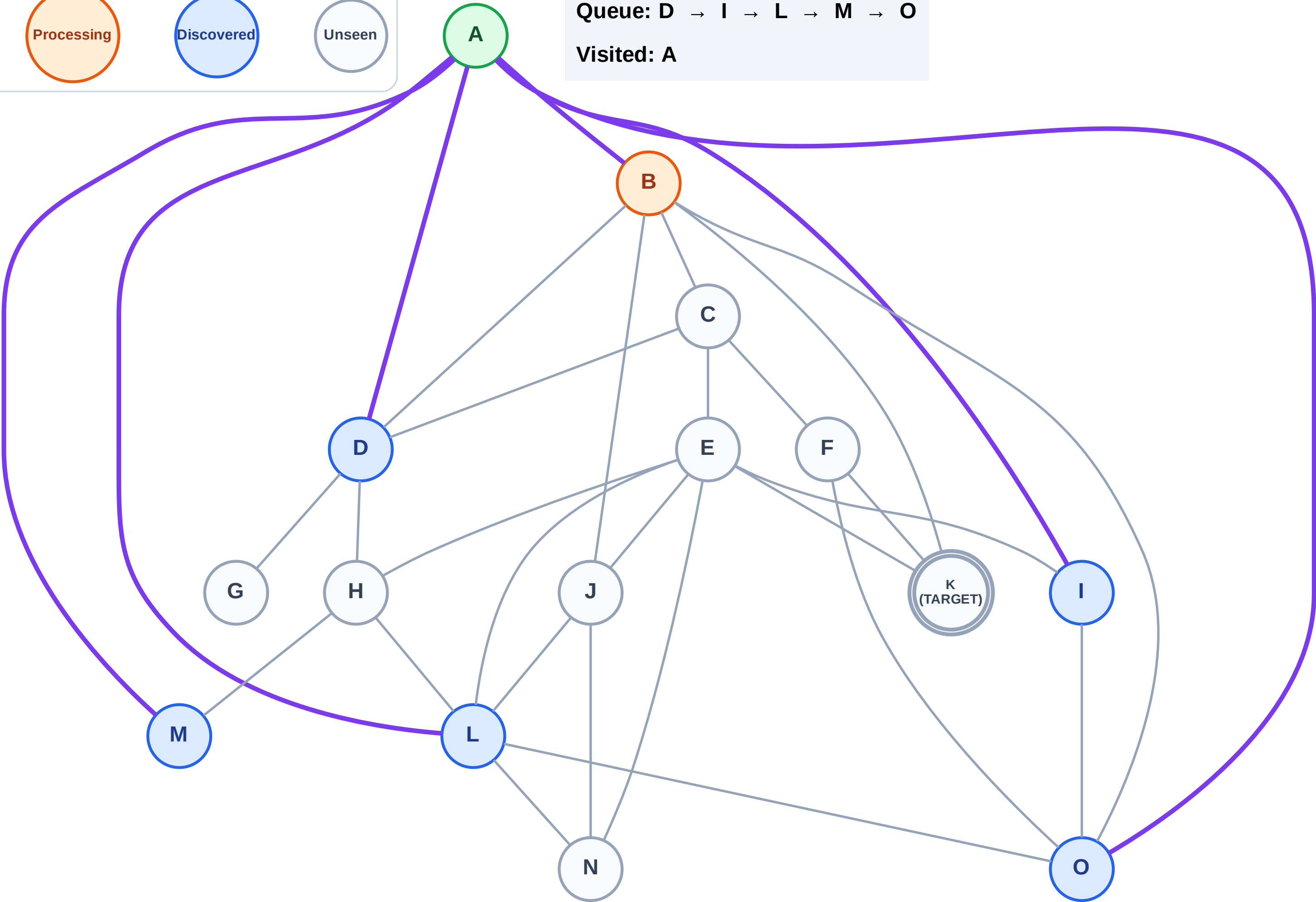
Processing

Discovered

Unseen

Queue: D → I → L → M → O

Visited: A



BFS Traversal

Step 5: Discover C, J, K from B

Legend

Complete

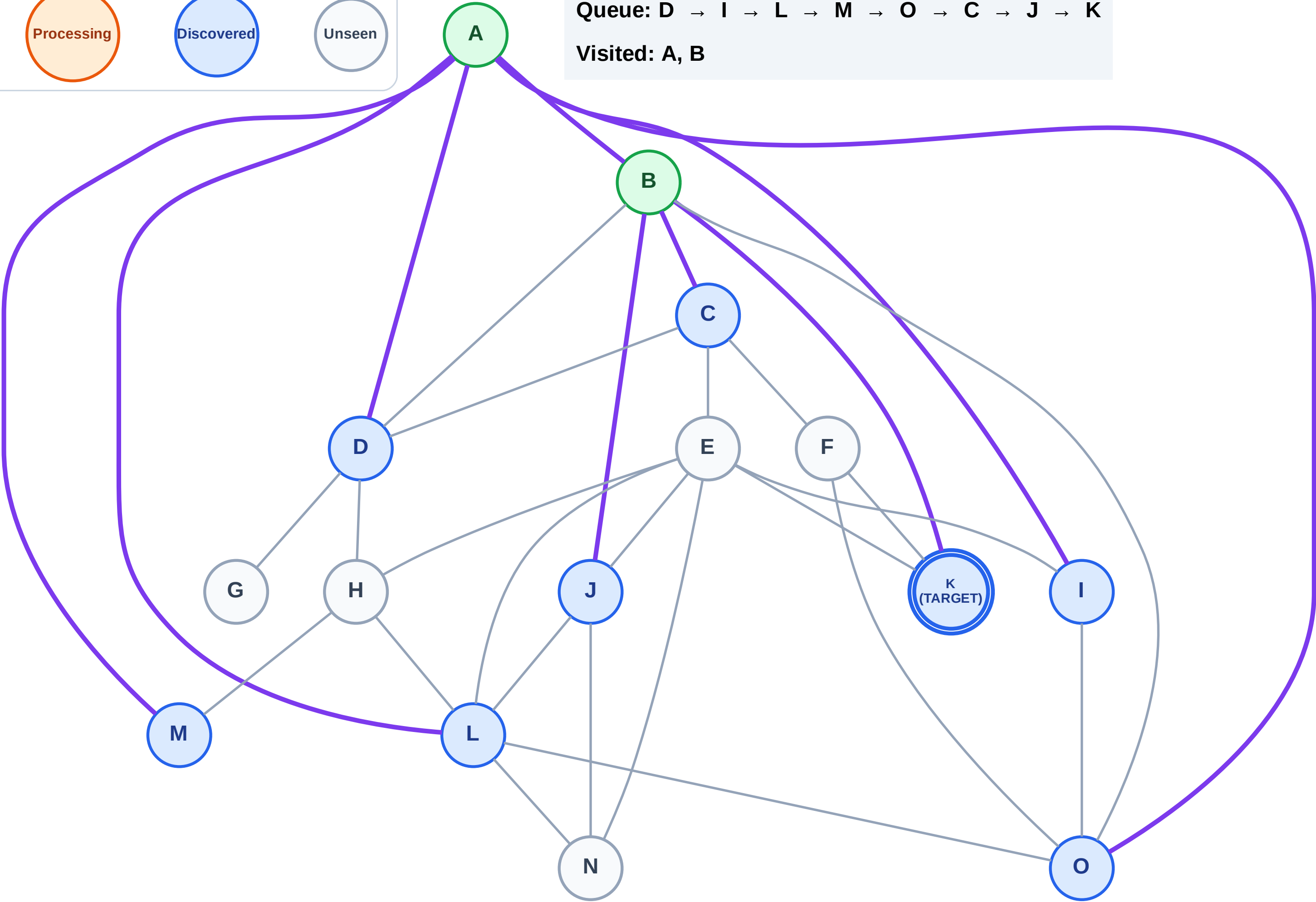
Processing

Discovered

Unseen

Queue: D → I → L → M → O → C → J → K

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

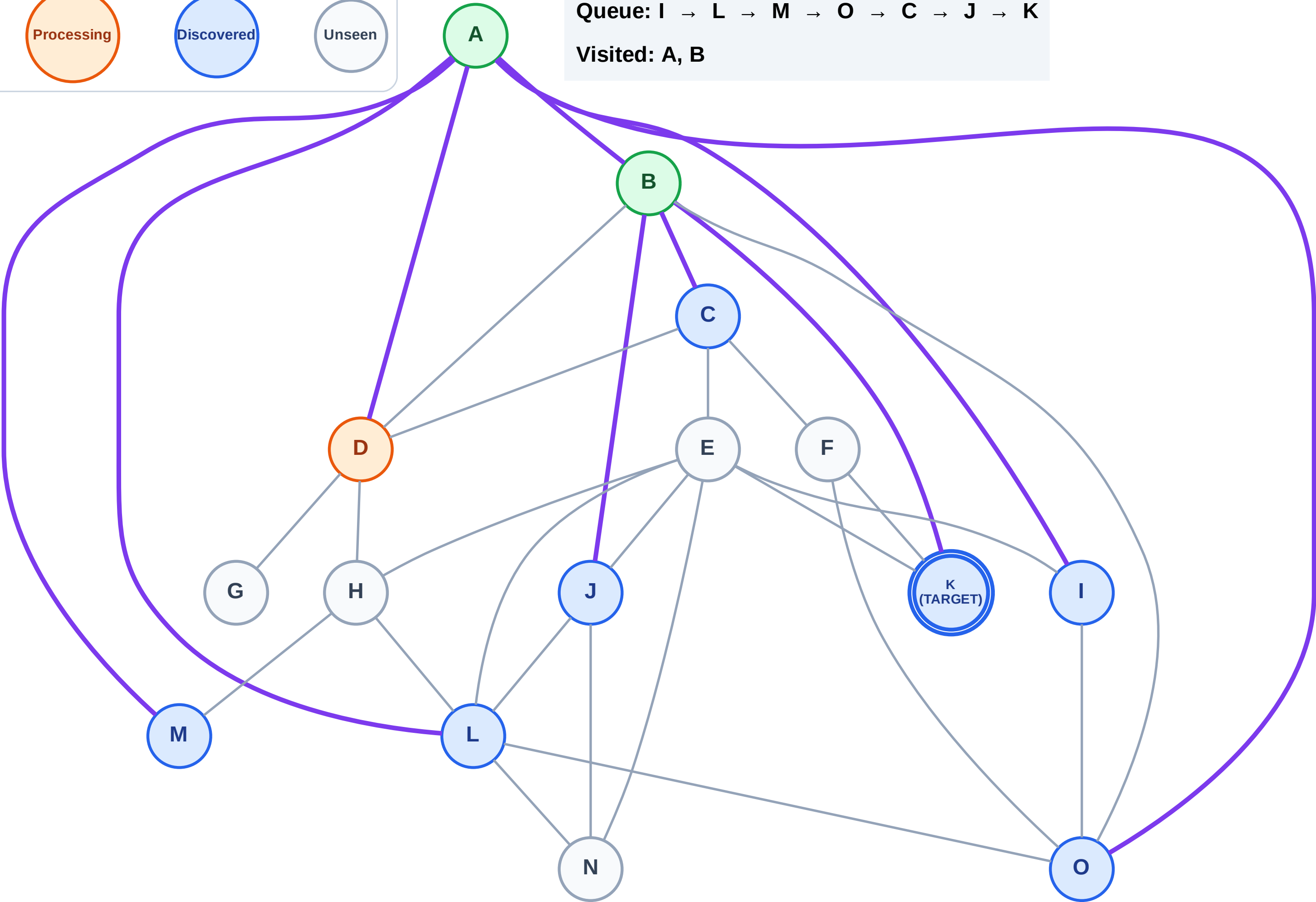
Processing

Discovered

Unseen

Queue: I → L → M → O → C → J → K

Visited: A, B



BFS Traversal

Step 7: Discover G, H from D

Legend

Complete

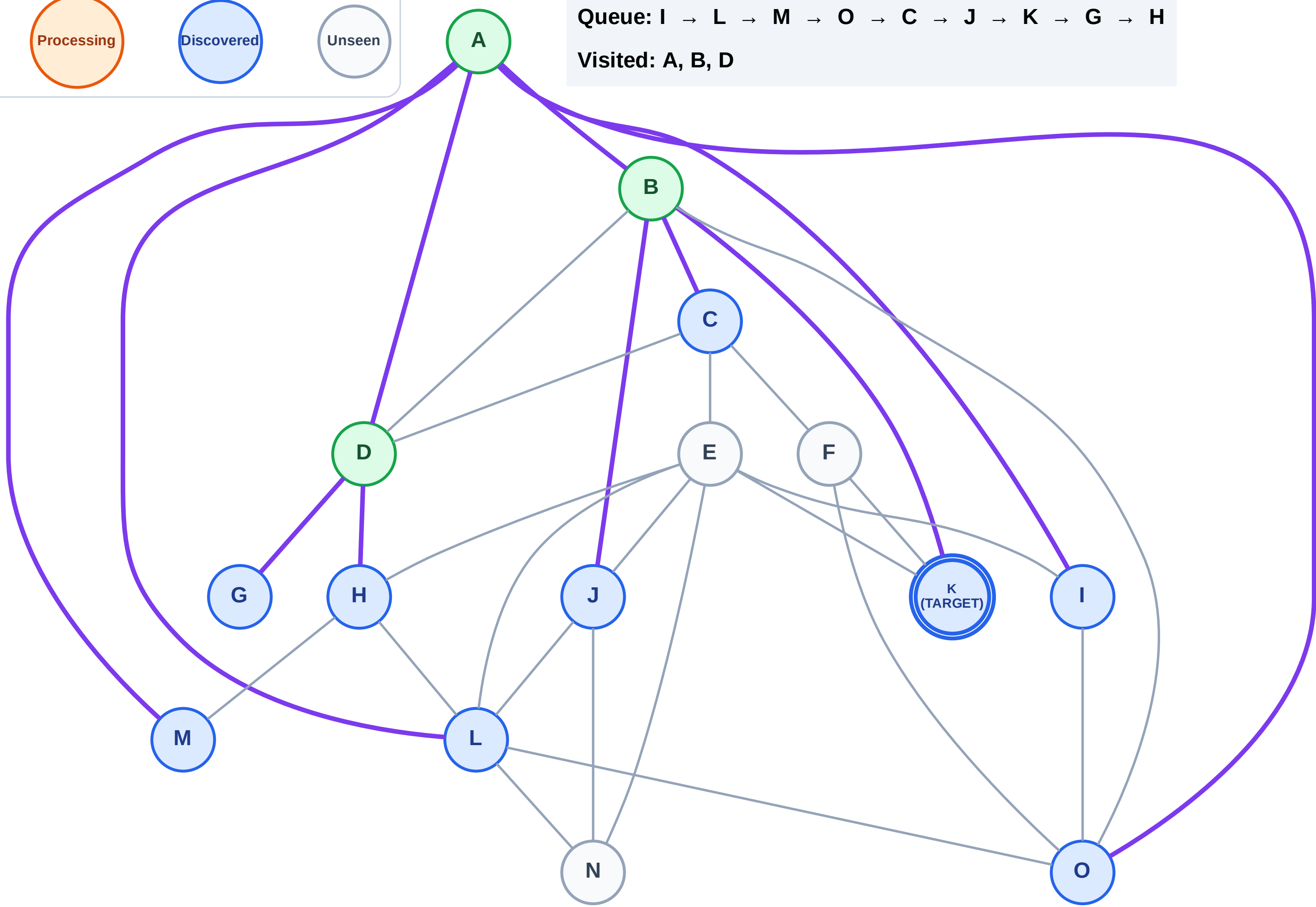
Processing

Discovered

Unseen

Queue: I → L → M → O → C → J → K → G → H

Visited: A, B, D



BFS Traversal

Step 8: Process node I

Legend

Complete

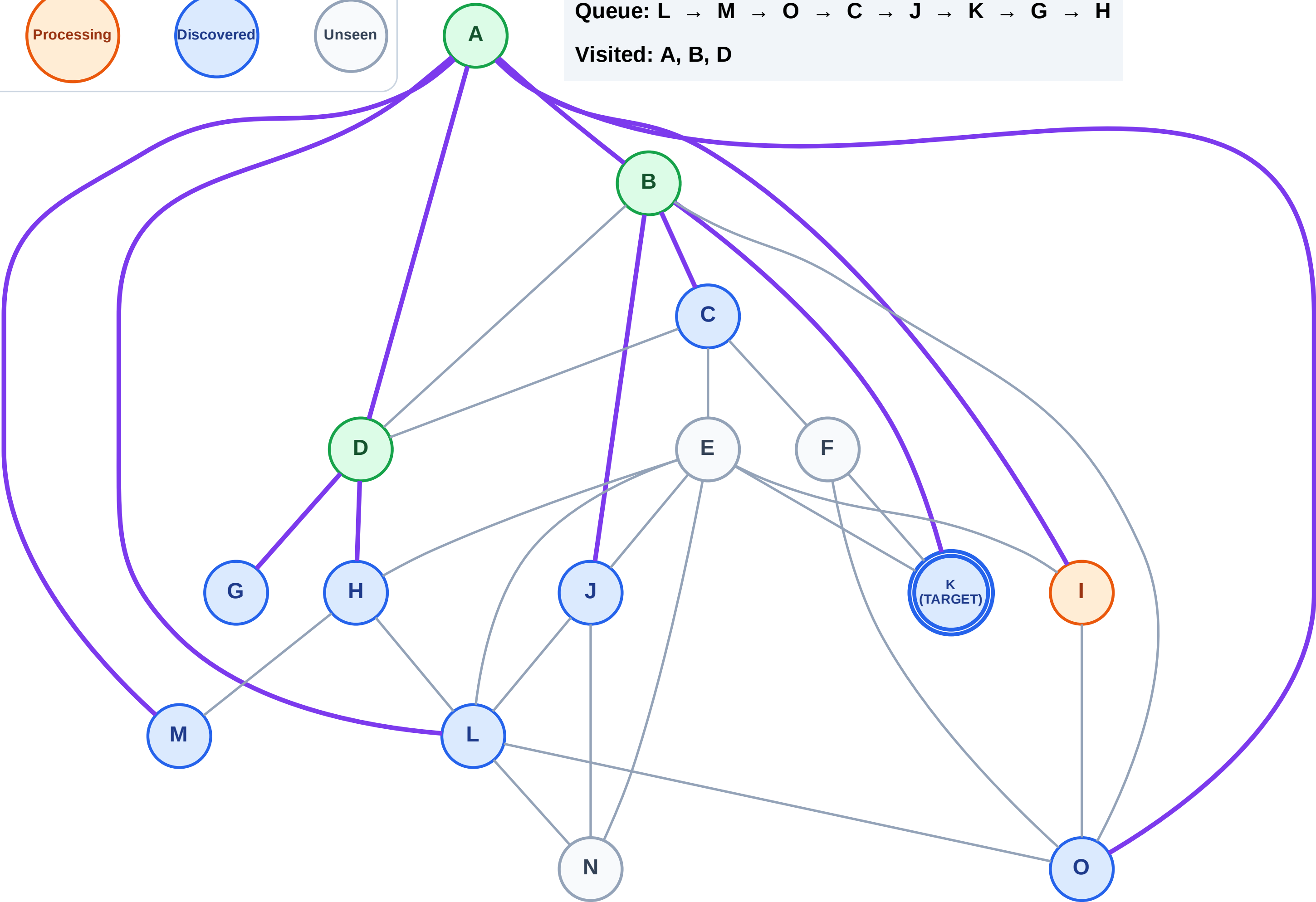
Processing

Discovered

Unseen

Queue: L → M → O → C → J → K → G → H

Visited: A, B, D



BFS Traversal

Step 9: Discover E from I

Legend

Complete

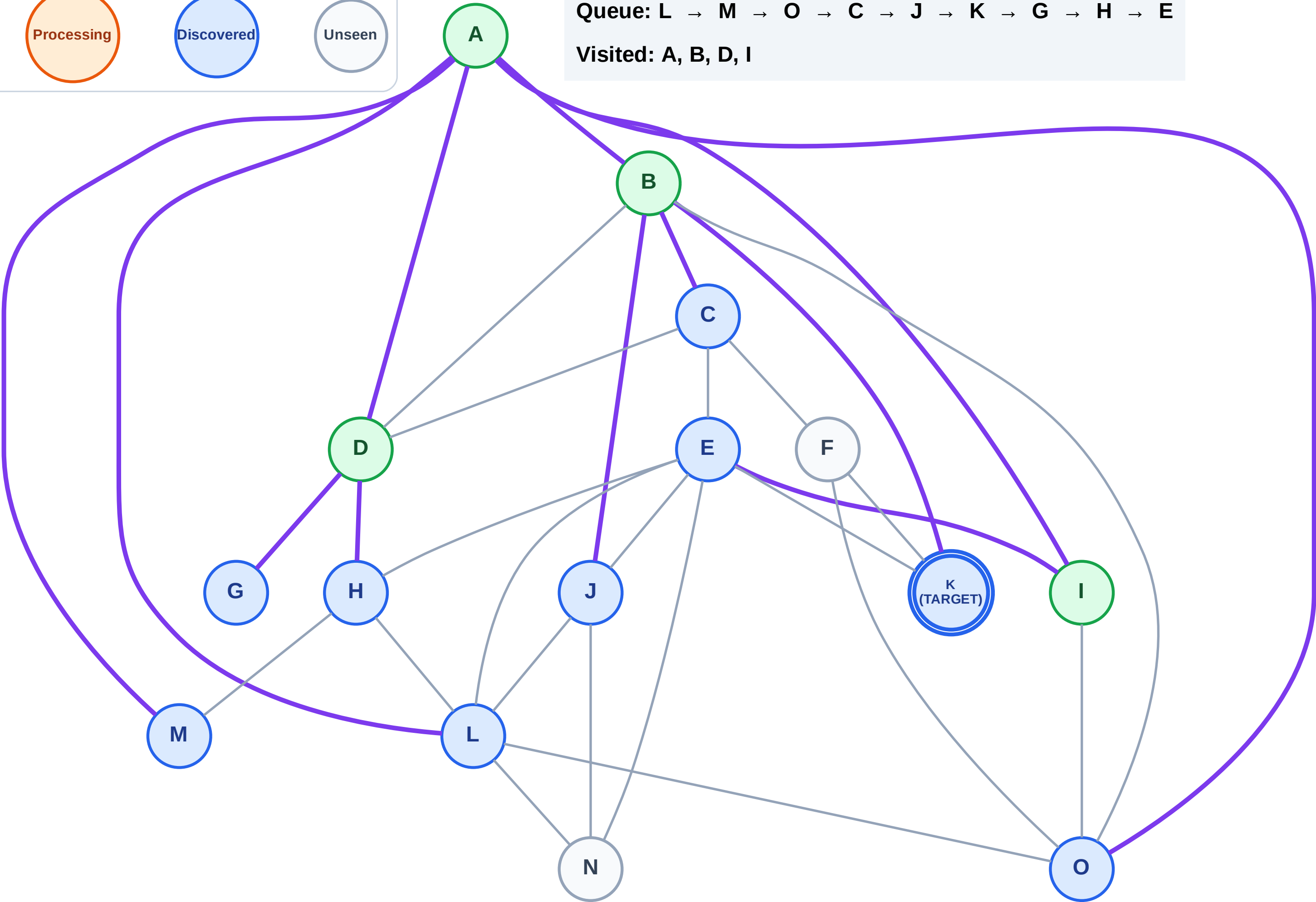
Processing

Discovered

Unseen

Queue: L → M → O → C → J → K → G → H → E

Visited: A, B, D, I



BFS Traversal

Step 10: Process node L

Legend

Complete

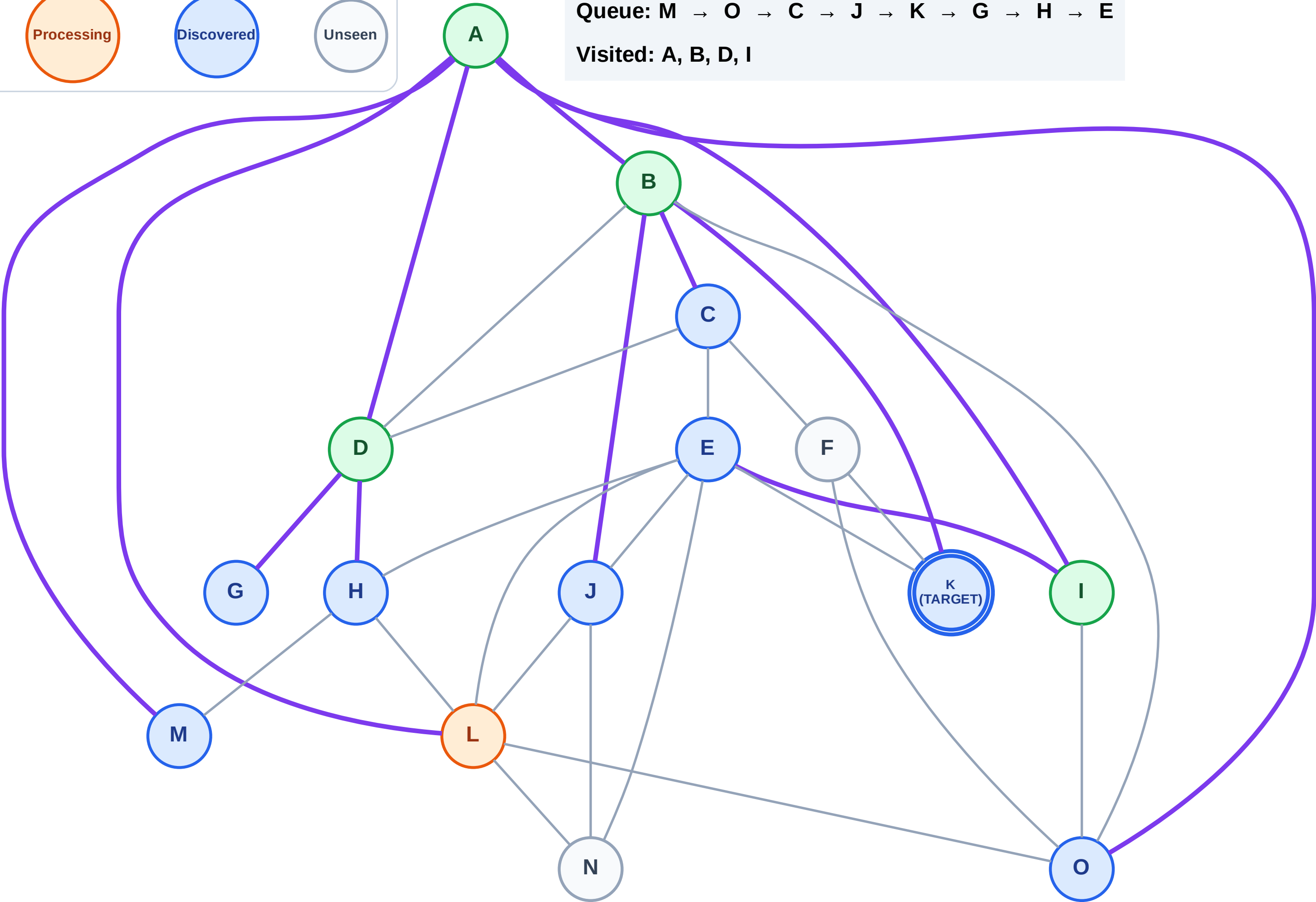
Processing

Discovered

Unseen

Queue: M → O → C → J → K → G → H → E

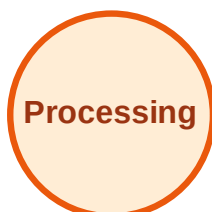
Visited: A, B, D, I



BFS Traversal

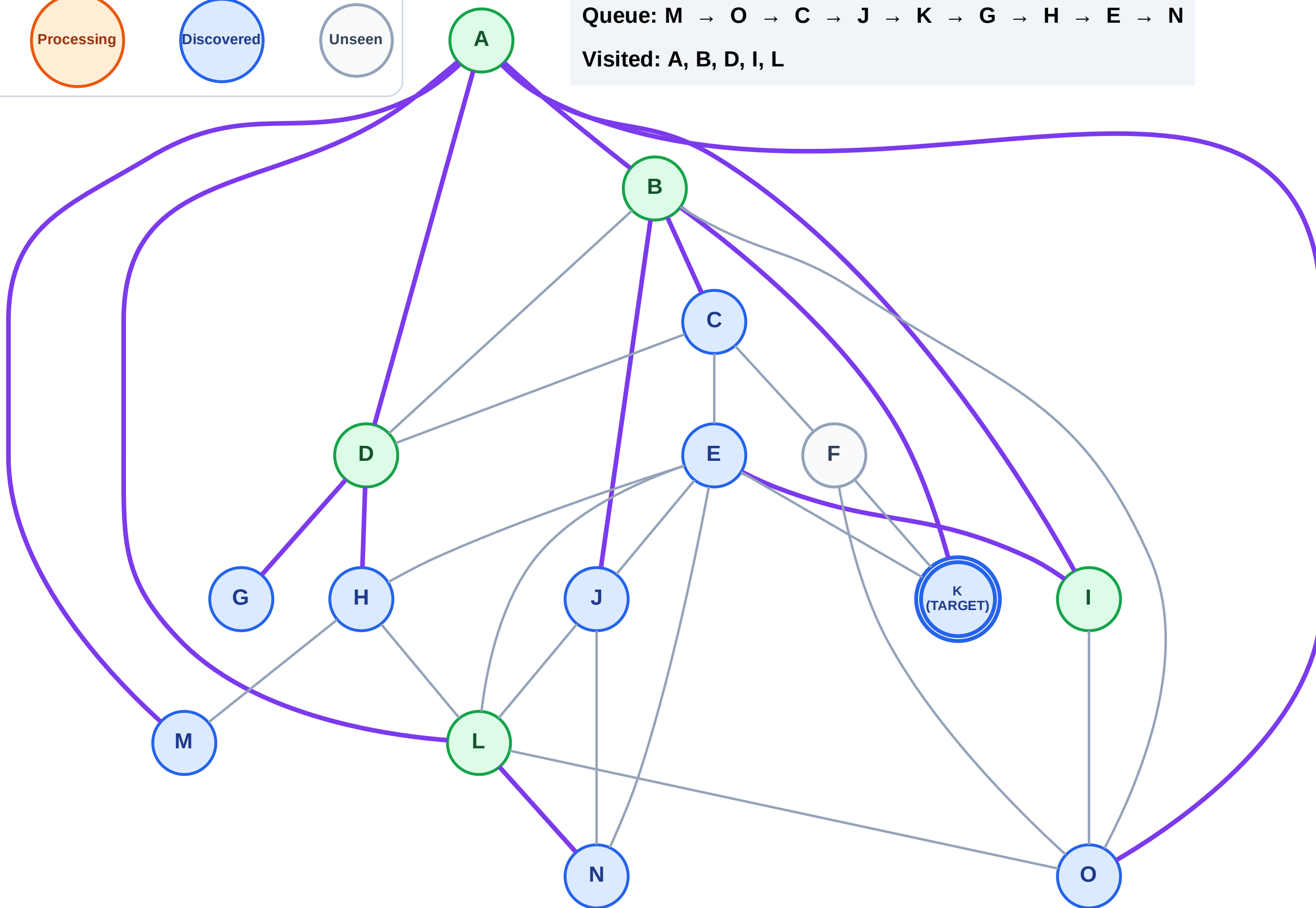
Step 11: Discover N from L

Legend



Queue: M → O → C → J → K → G → H → E → N

Visited: A, B, D, I, L



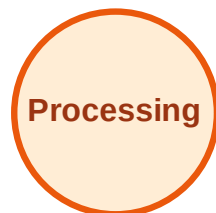
BFS Traversal

Step 12: Process node M

Legend



Complete



Processing



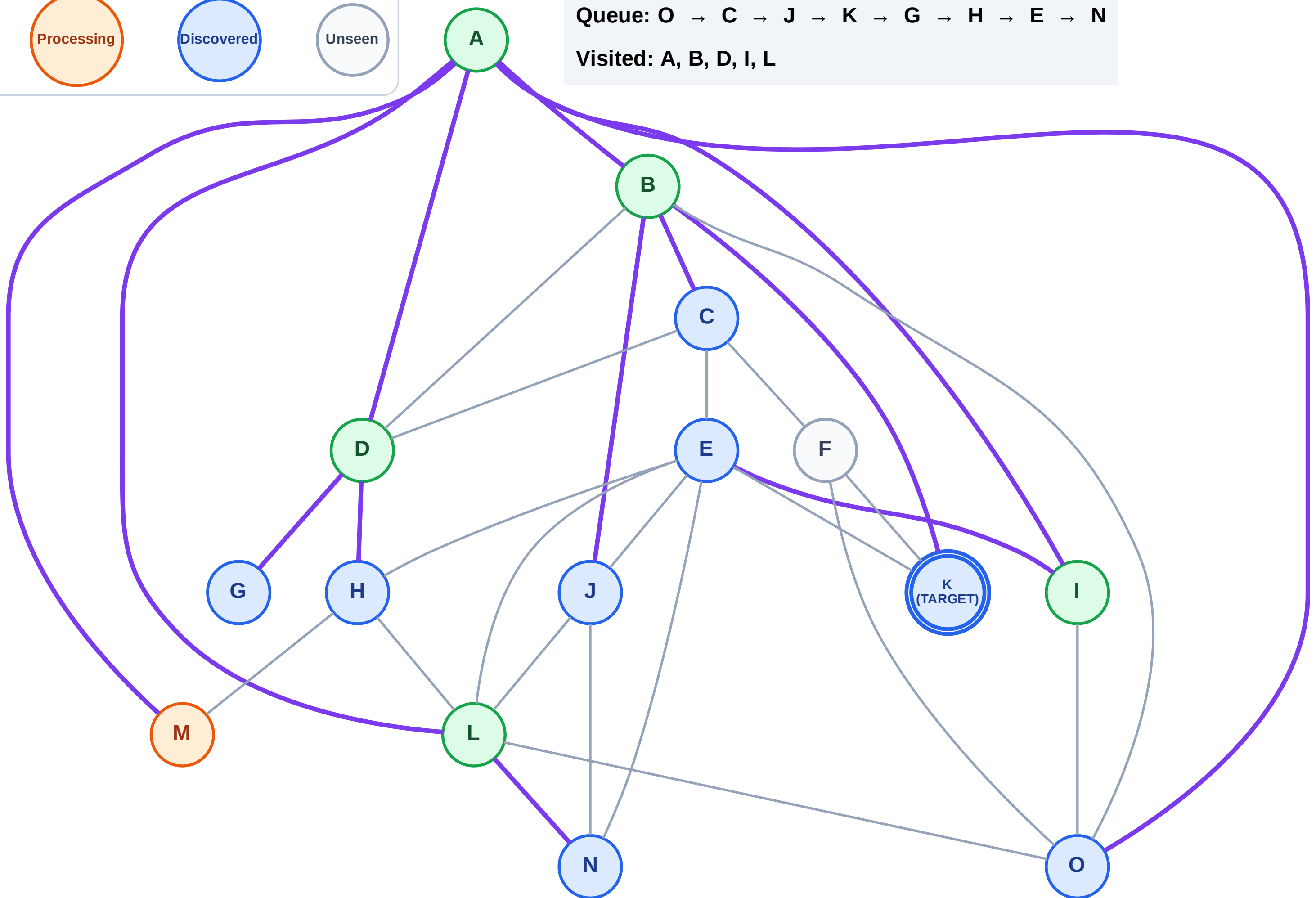
Discovered



Unseen

Queue: O → C → J → K → G → H → E → N

Visited: A, B, D, I, L



BFS Traversal

Step 13: No new neighbors from M

Legend

Complete

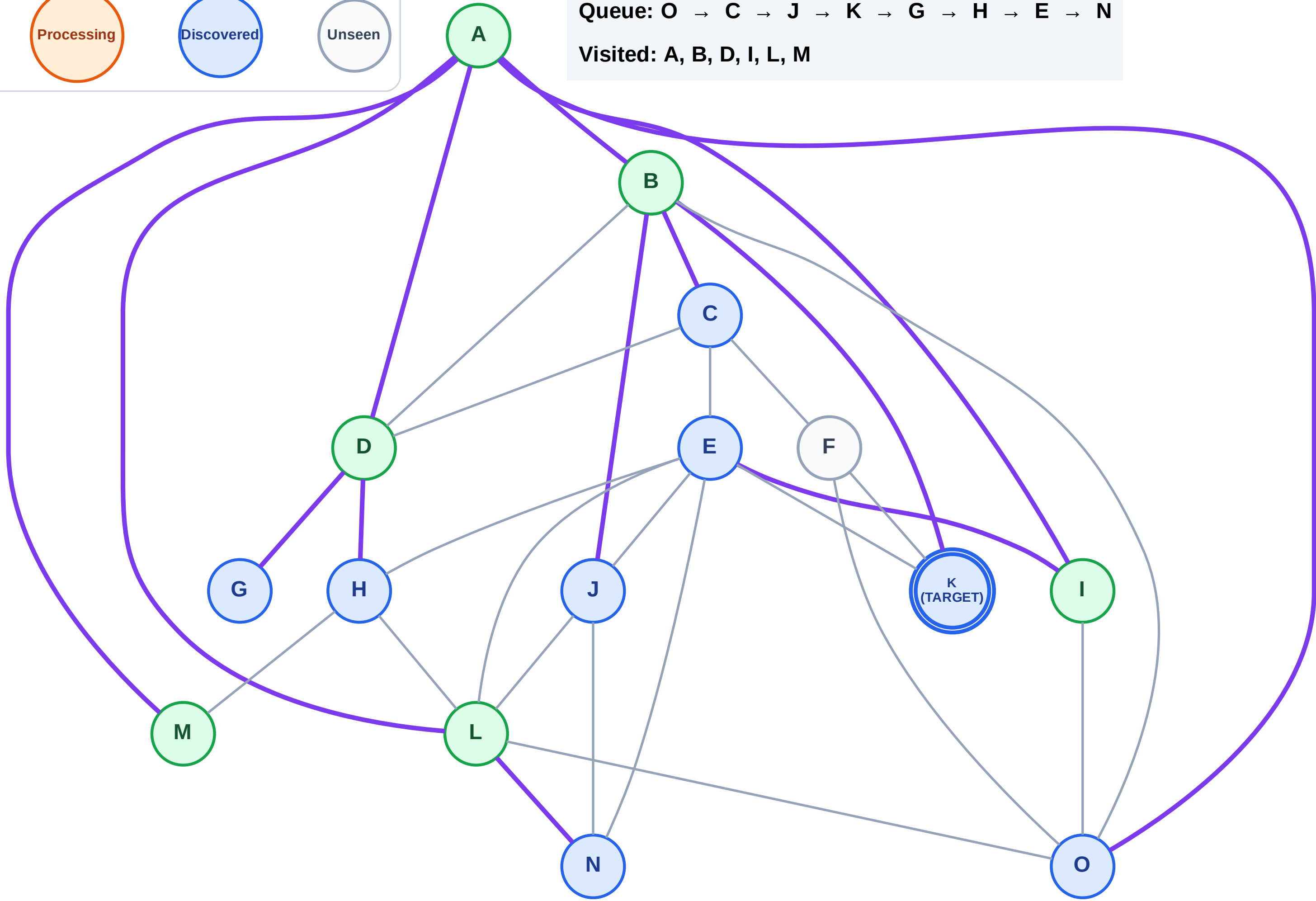
Processing

Discovered

Unseen

Queue: O → C → J → K → G → H → E → N

Visited: A, B, D, I, L, M



BFS Traversal

Step 14: Process node O

Legend

Complete

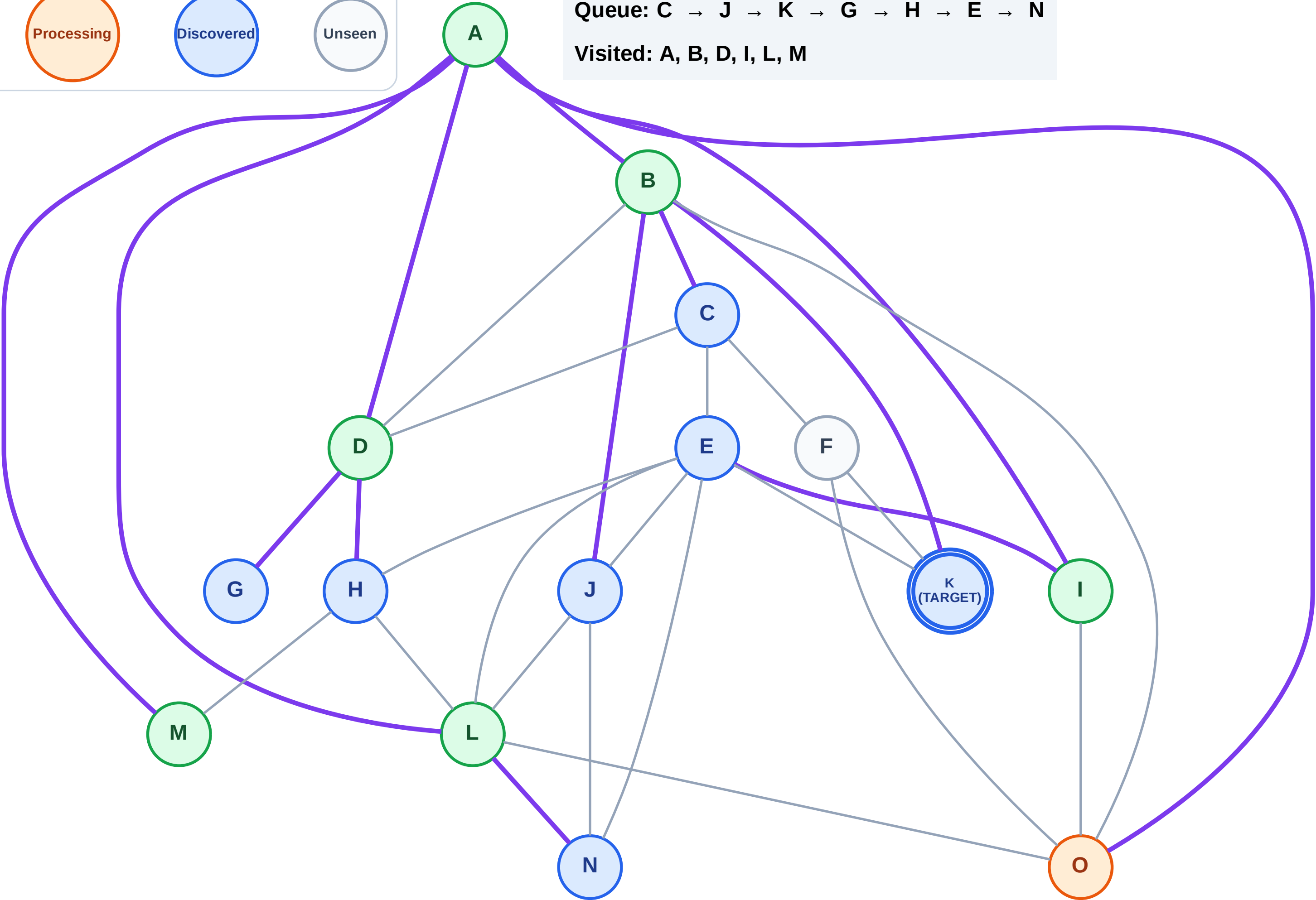
Processing

Discovered

Unseen

Queue: C → J → K → G → H → E → N

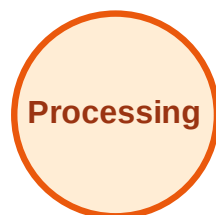
Visited: A, B, D, I, L, M



BFS Traversal

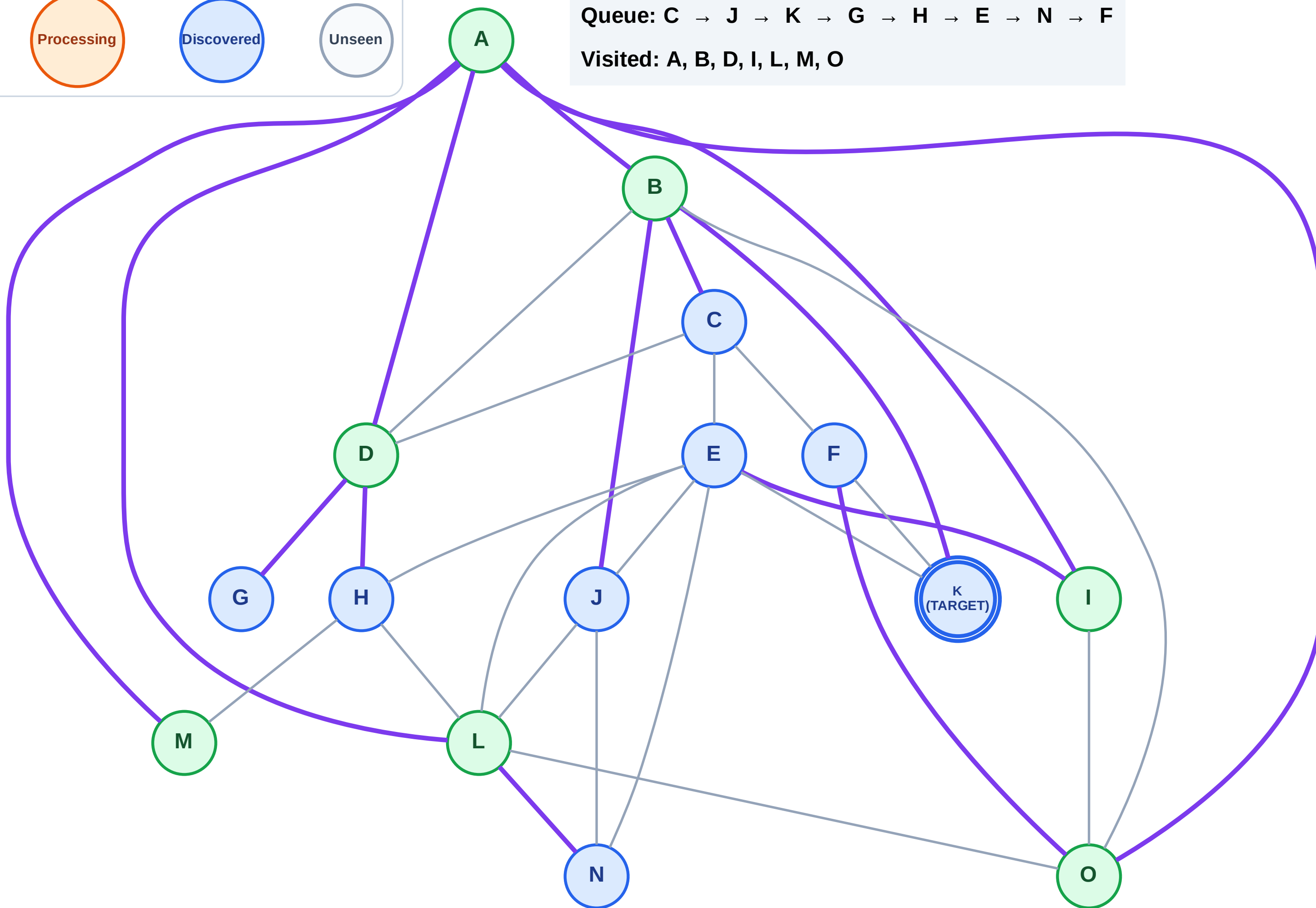
Step 15: Discover F from O

Legend



Queue: C → J → K → G → H → E → N → F

Visited: A, B, D, I, L, M, O



BFS Traversal

Step 16: Process node C

Legend

Complete

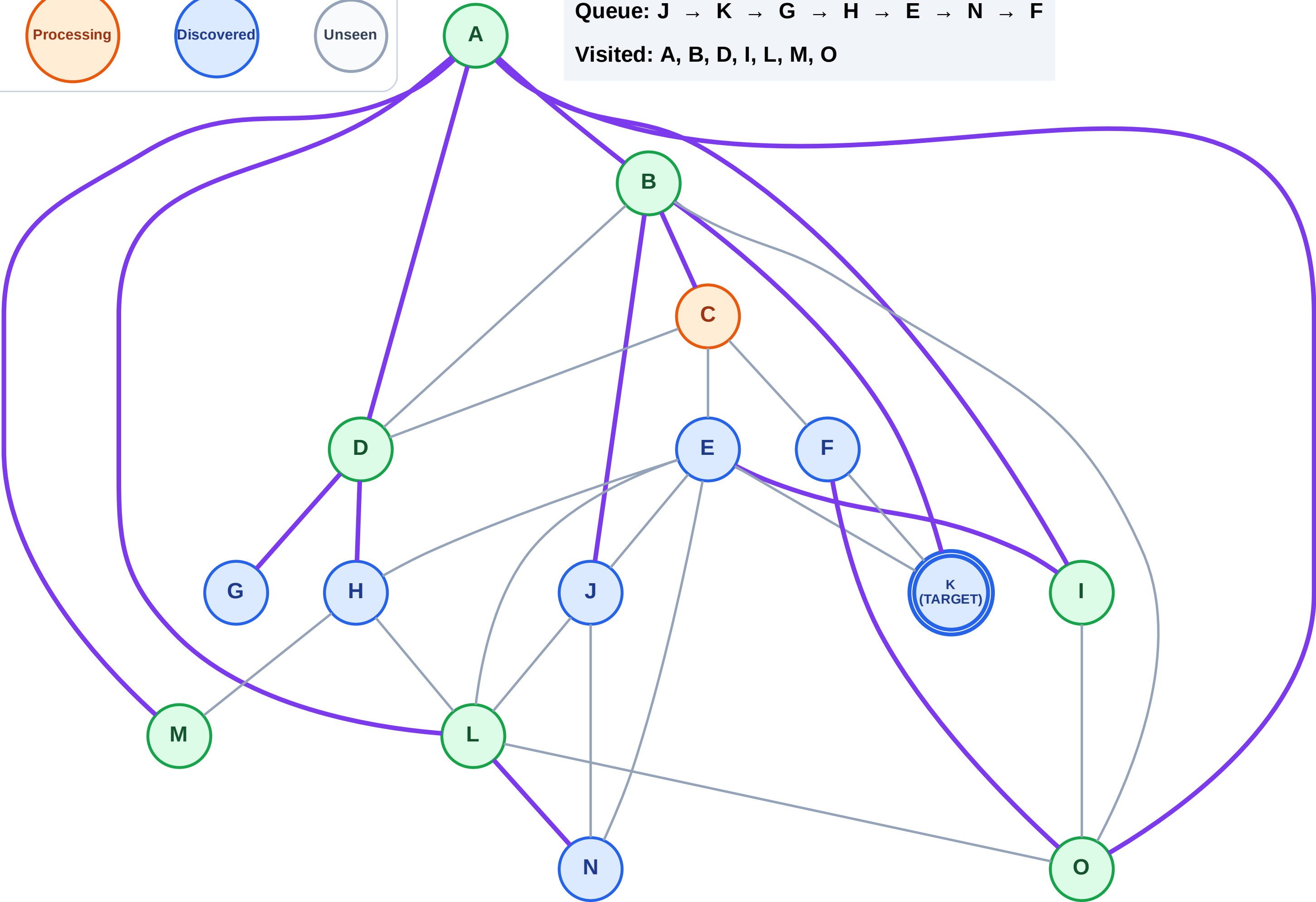
Processing

Discovered

Unseen

Queue: J → K → G → H → E → N → F

Visited: A, B, D, I, L, M, O



BFS Traversal

Step 17: No new neighbors from C

Legend

Complete

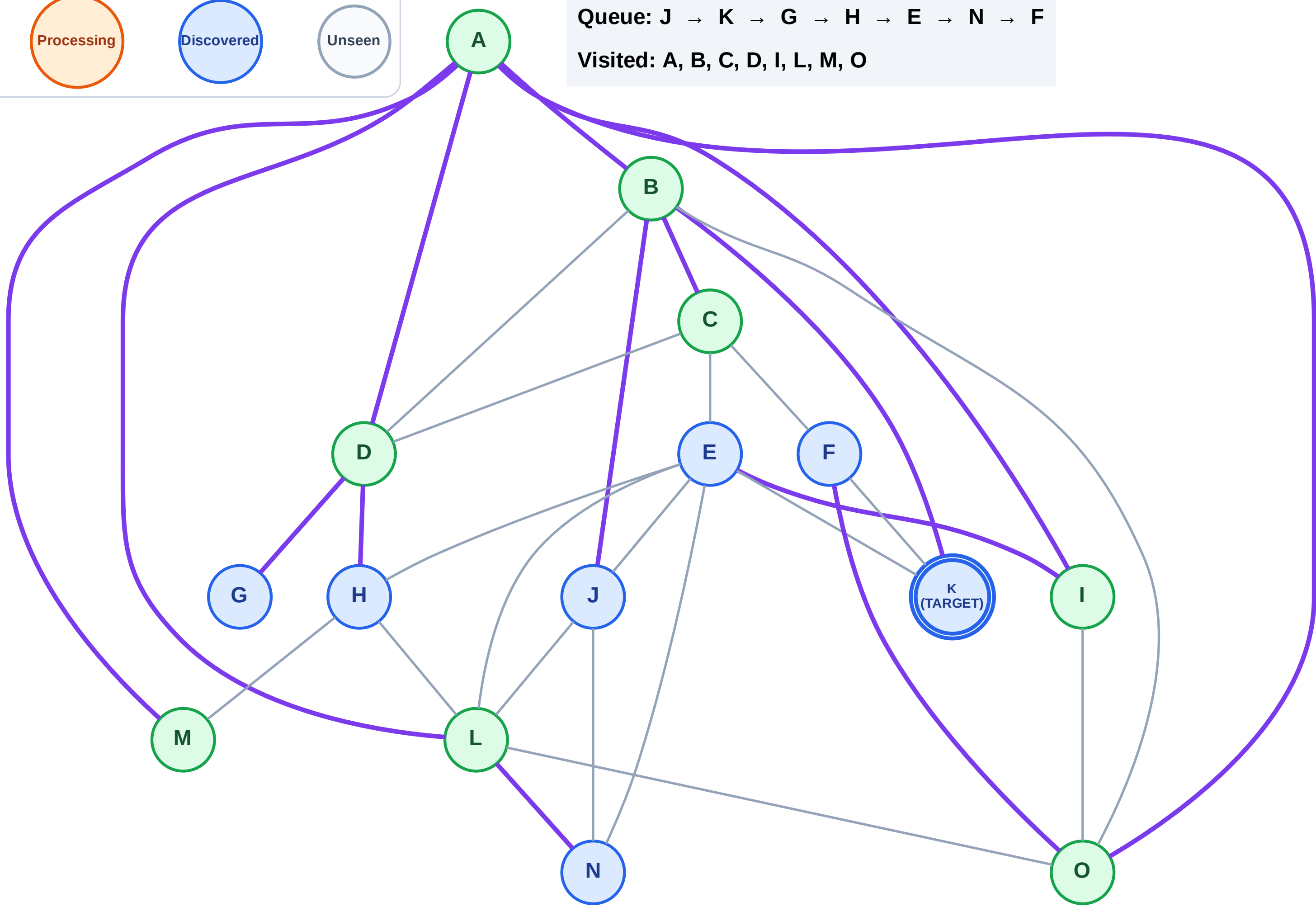
Processing

Discovered

Unseen

Queue: J → K → G → H → E → N → F

Visited: A, B, C, D, I, L, M, O



BFS Traversal

Step 18: Process node J

Legend

Complete

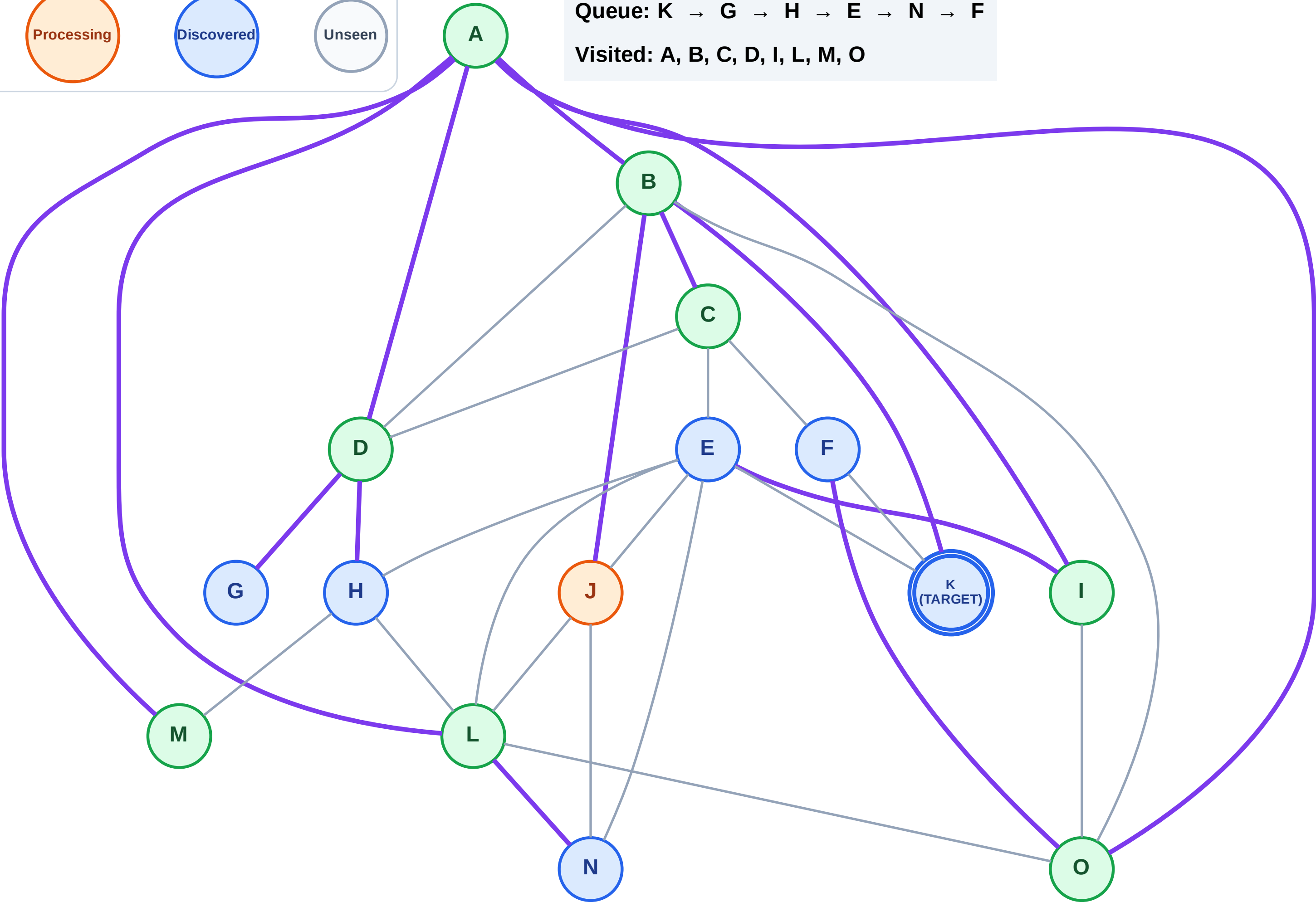
Processing

Discovered

Unseen

Queue: K → G → H → E → N → F

Visited: A, B, C, D, I, L, M, O



BFS Traversal

Step 19: No new neighbors from J

Legend

Complete

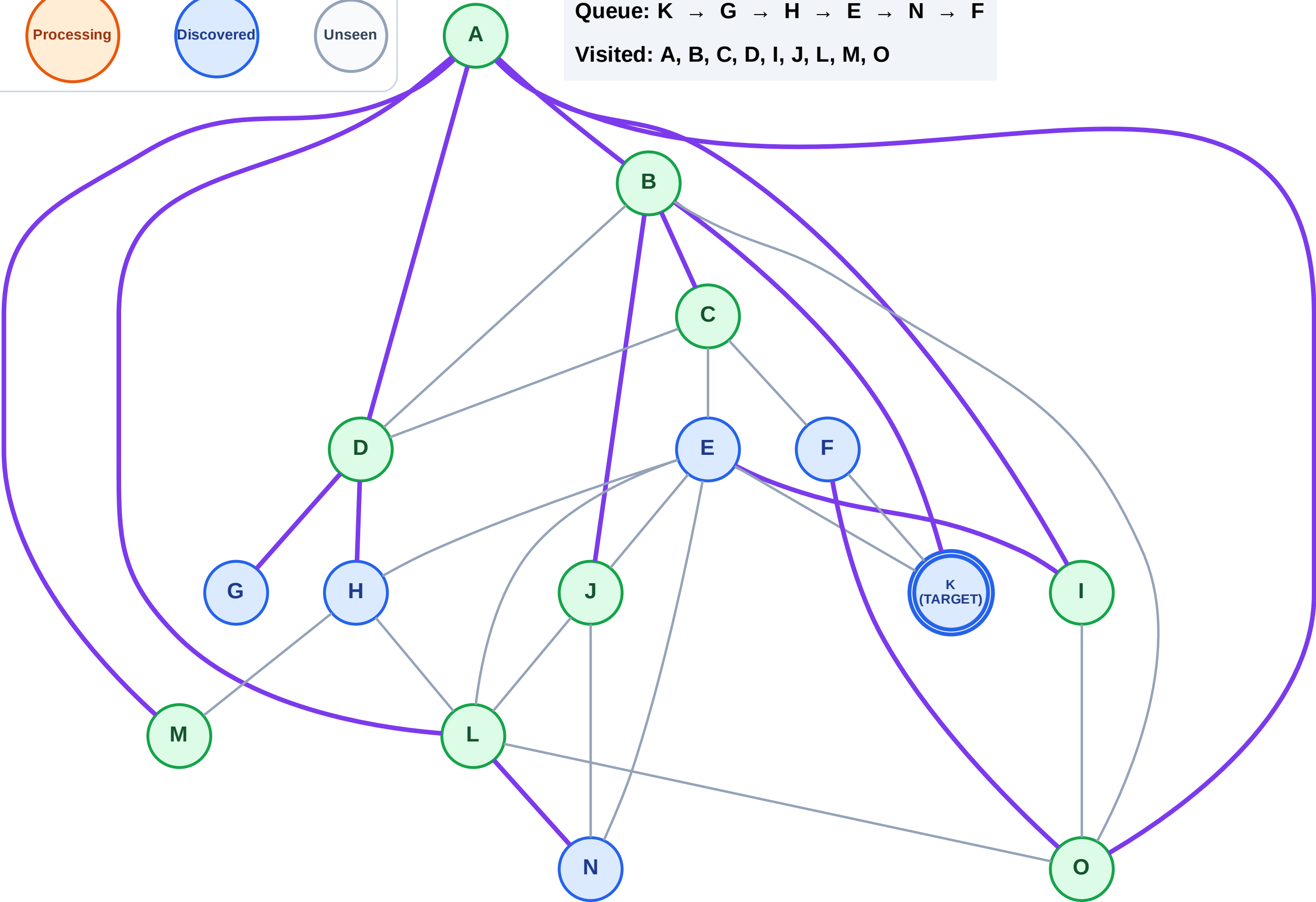
Processing

Discovered

Unseen

Queue: K → G → H → E → N → F

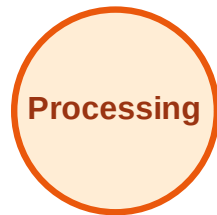
Visited: A, B, C, D, I, J, L, M, O



BFS Traversal

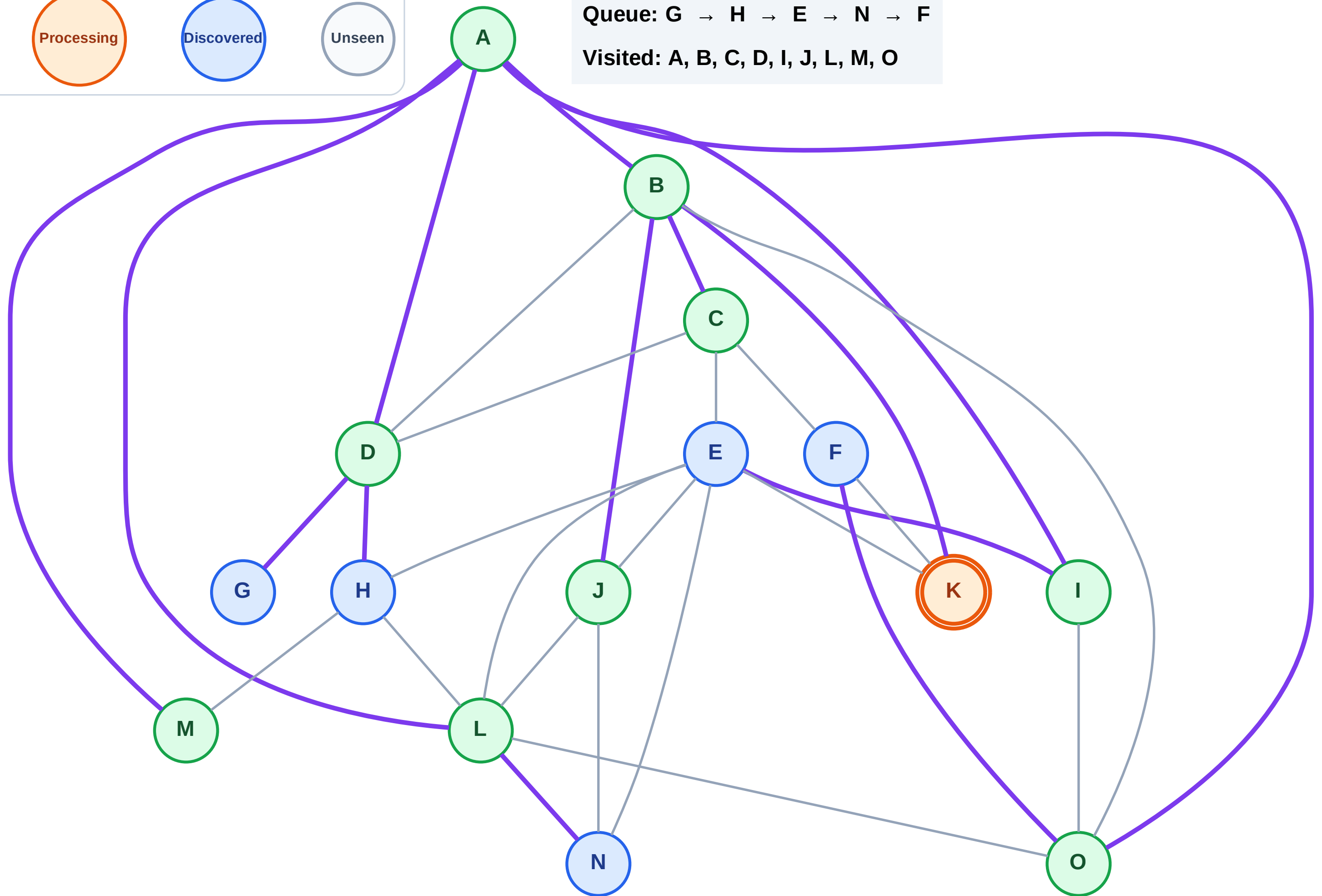
Step 20: Process node K

Legend



Queue: G → H → E → N → F

Visited: A, B, C, D, I, J, L, M, O



BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

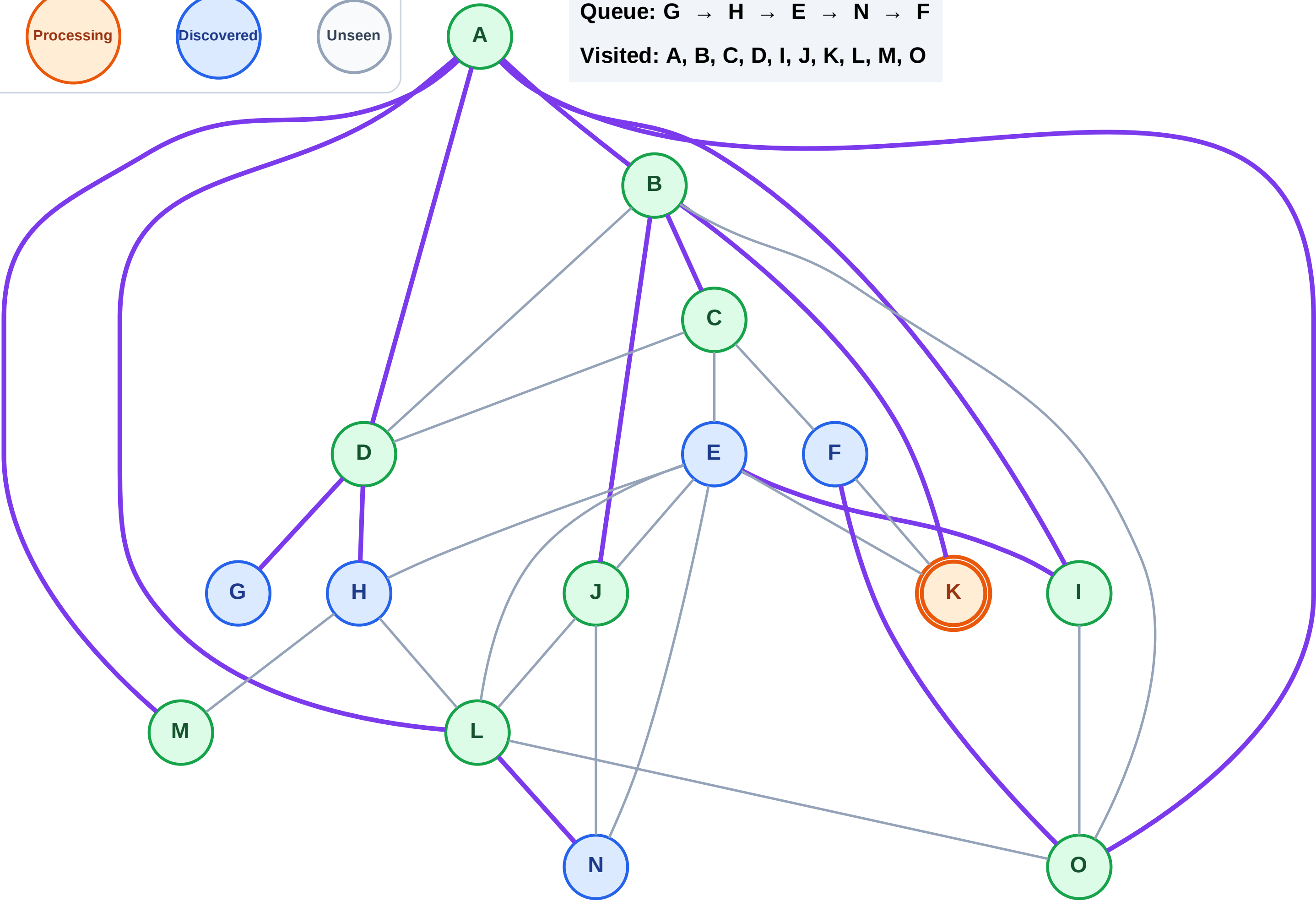
Processing

Discovered

Unseen

Queue: G → H → E → N → F

Visited: A, B, C, D, I, J, K, L, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

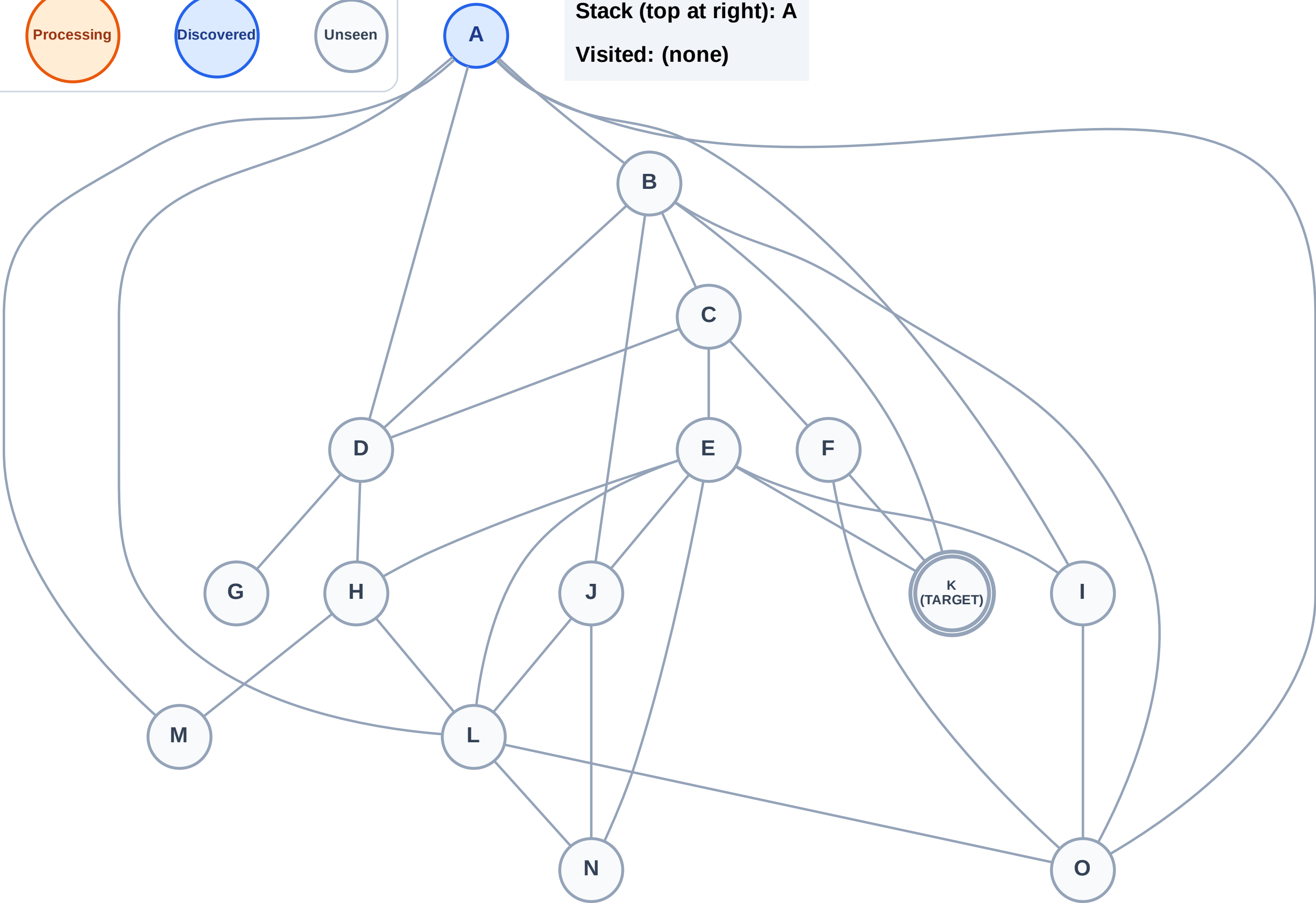
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

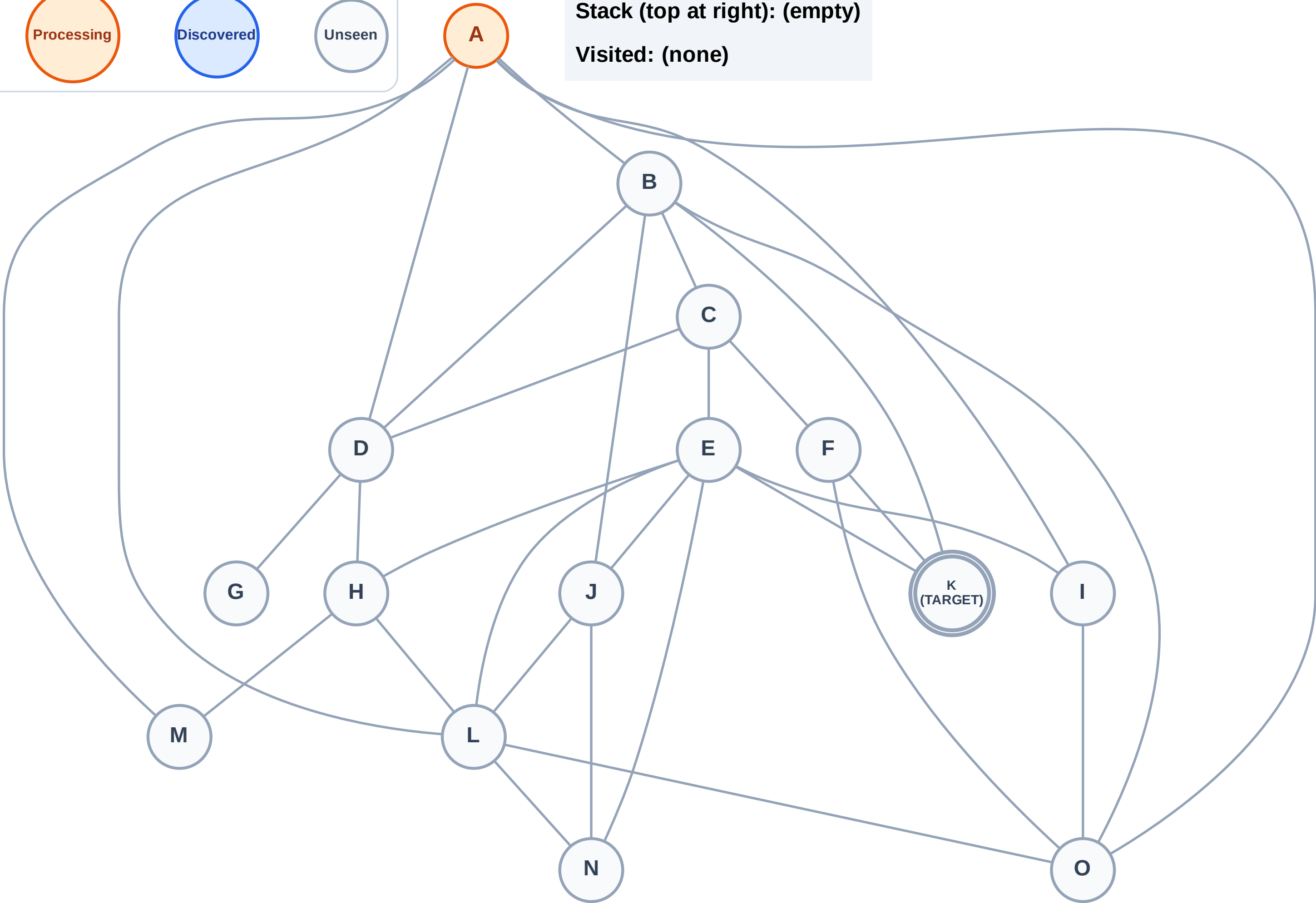
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, M, L, I, D, B from A

Legend

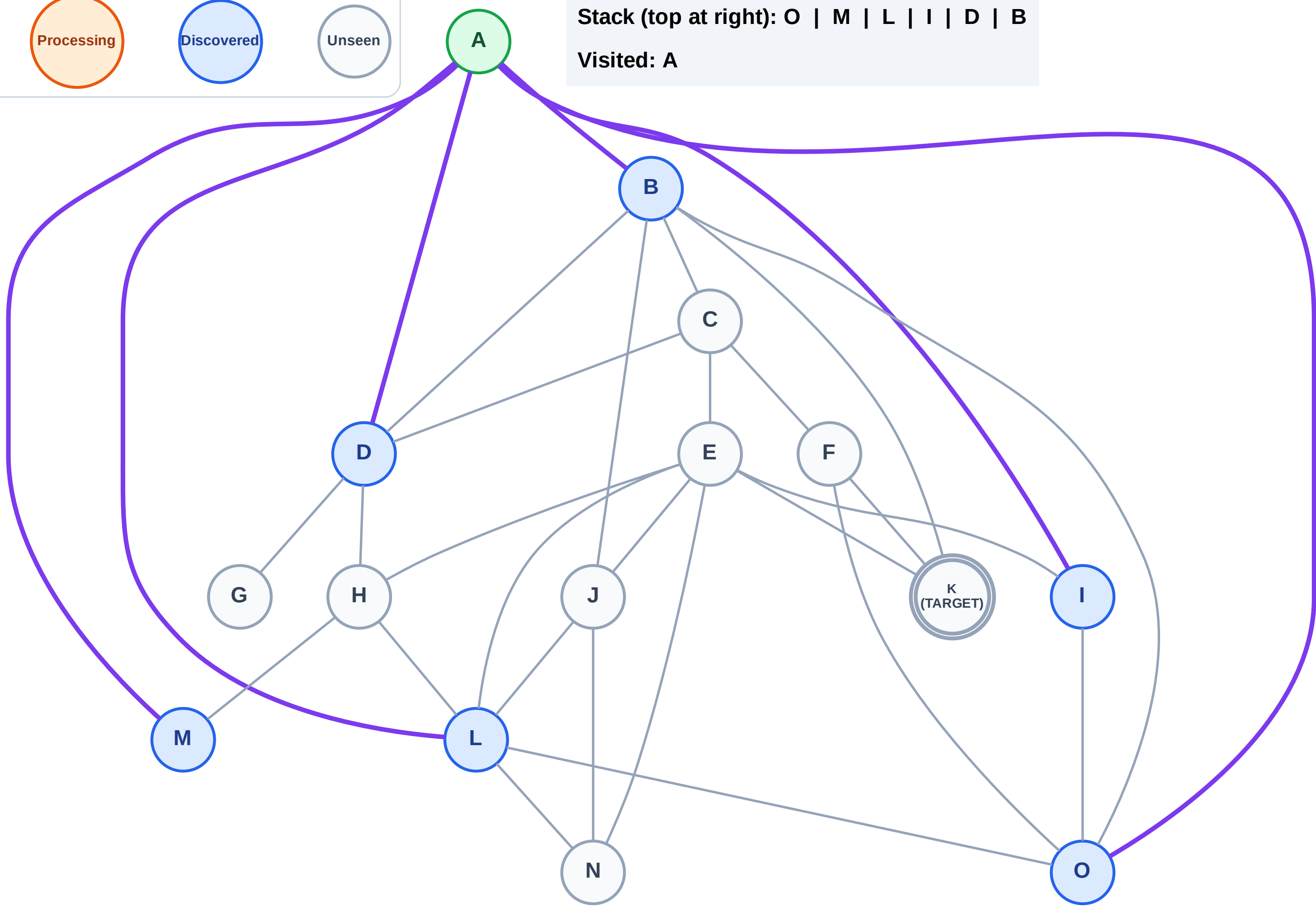
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

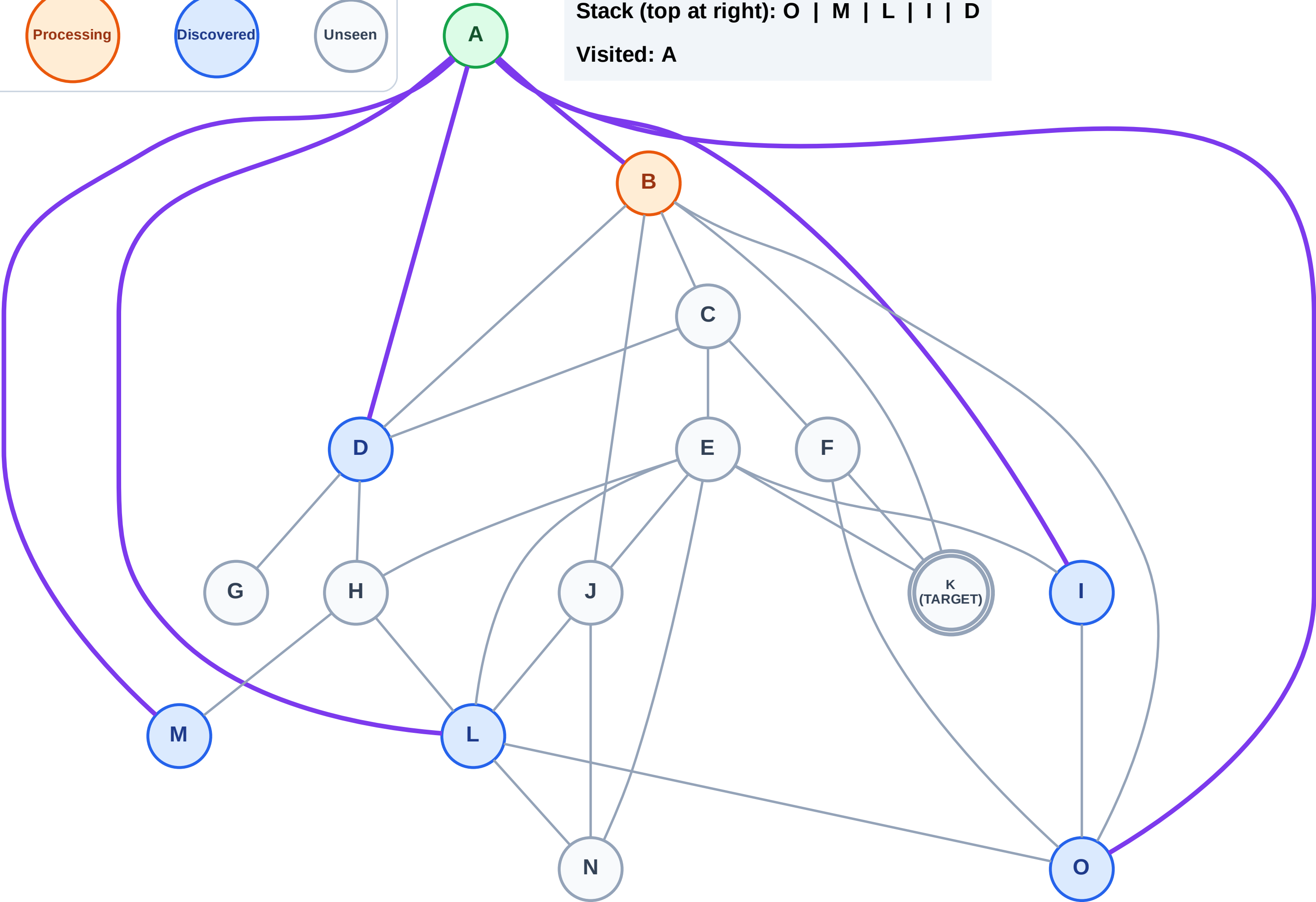
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D

Visited: A



DFS Traversal

Step 5: Discover K, J, C from B

Legend

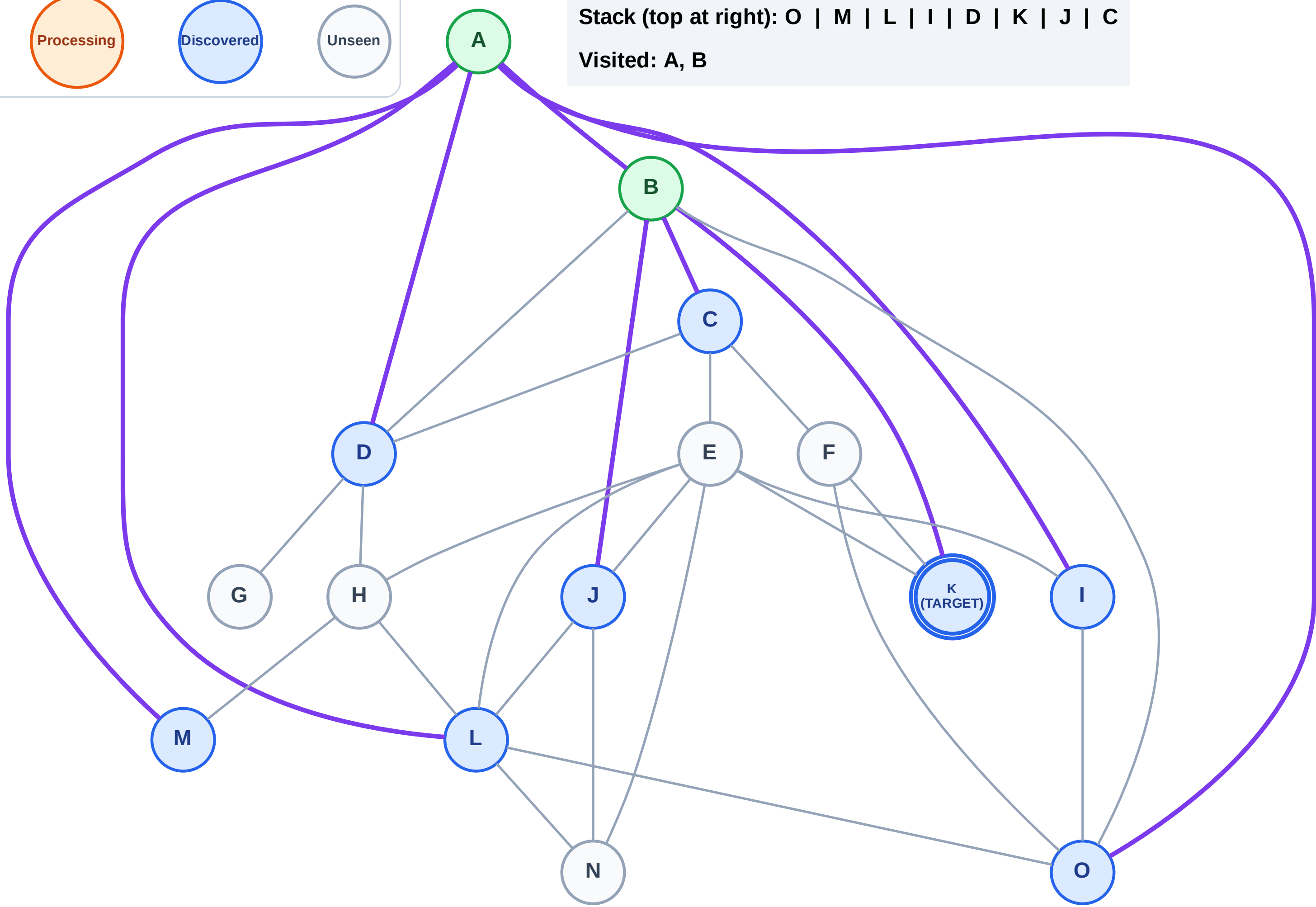
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J | C
Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

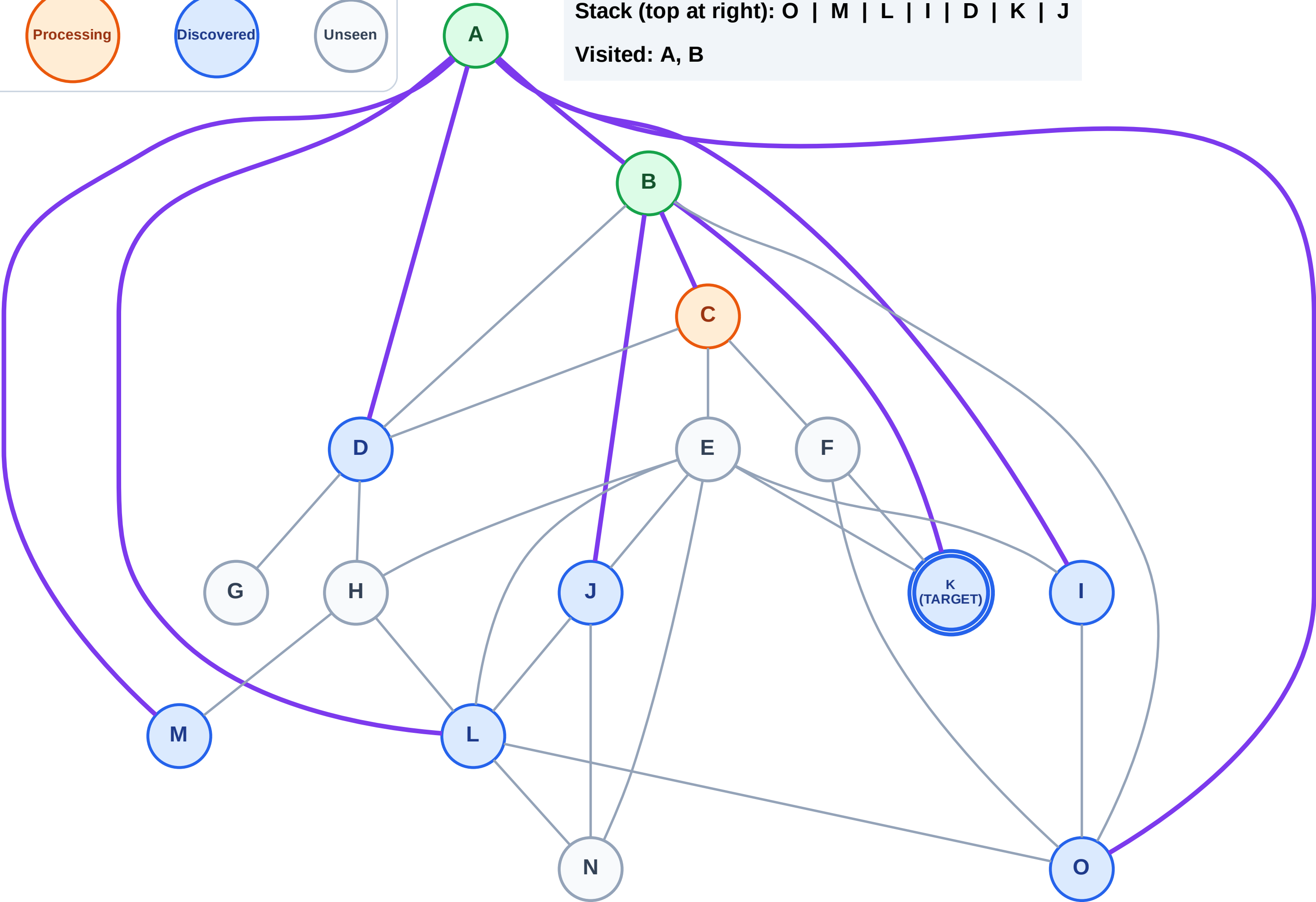
Processing

Discovered

Unseen

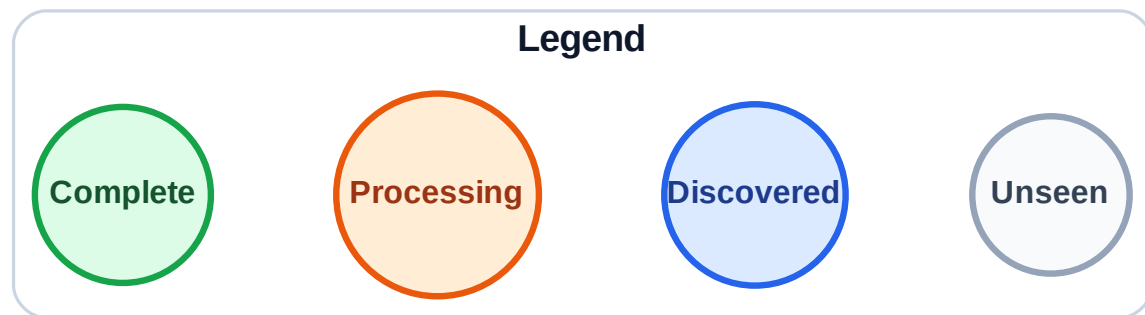
Stack (top at right): O | M | L | I | D | K | J

Visited: A, B



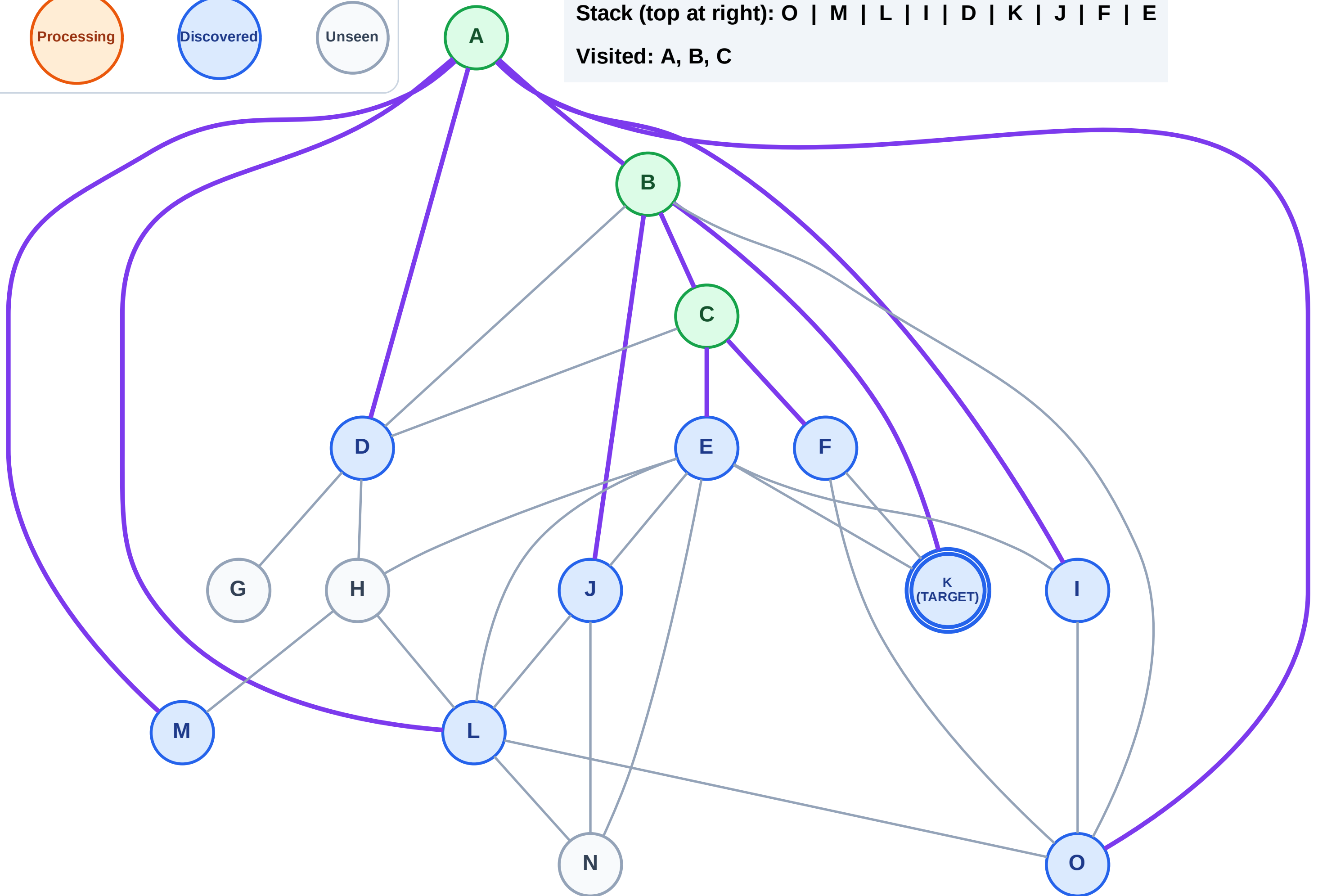
Step 7: Discover F, E from C

Step 7: Discover F, E from C



Stack (top at right): O | M | L | I | D | K | J | F | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

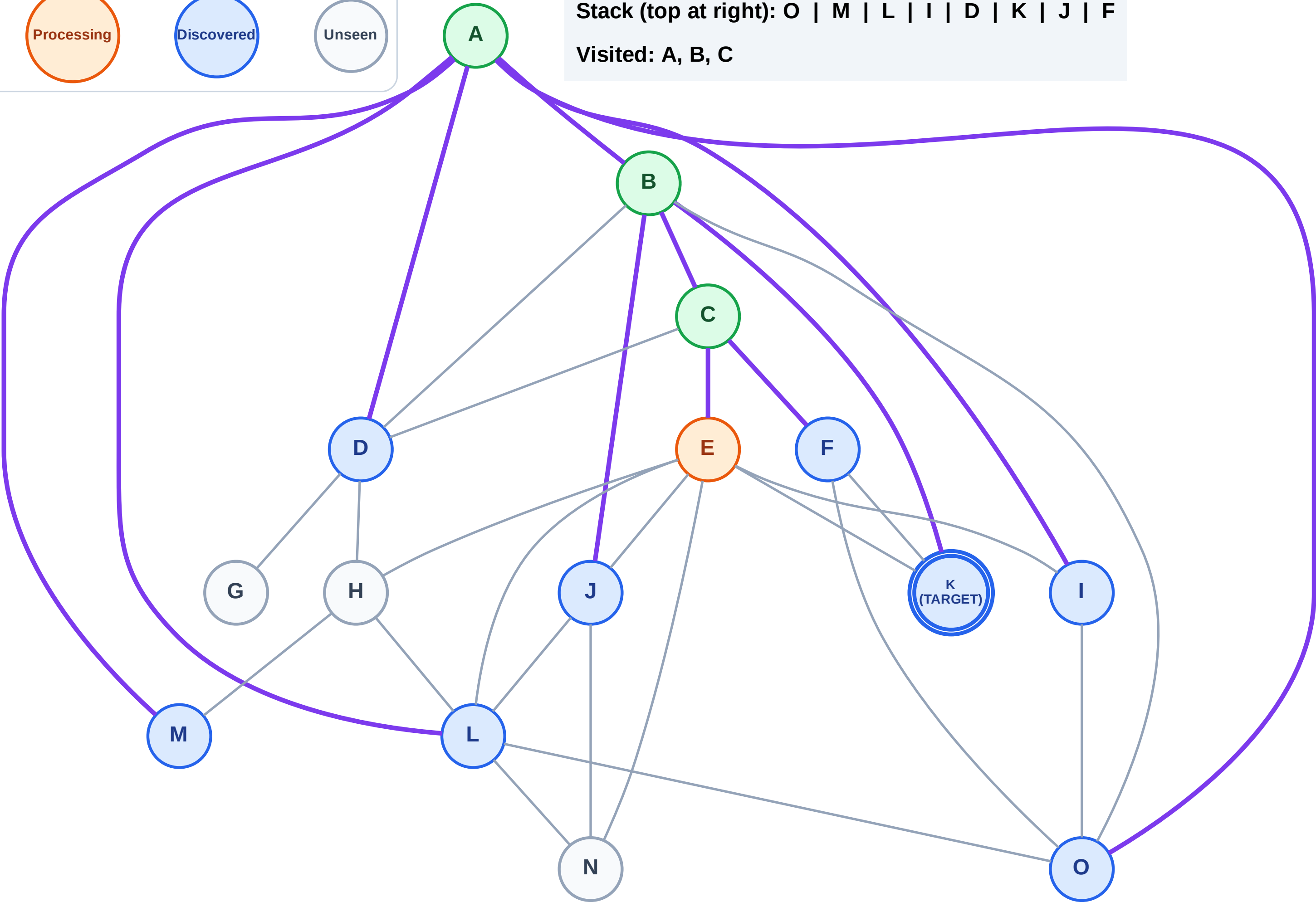
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J | F

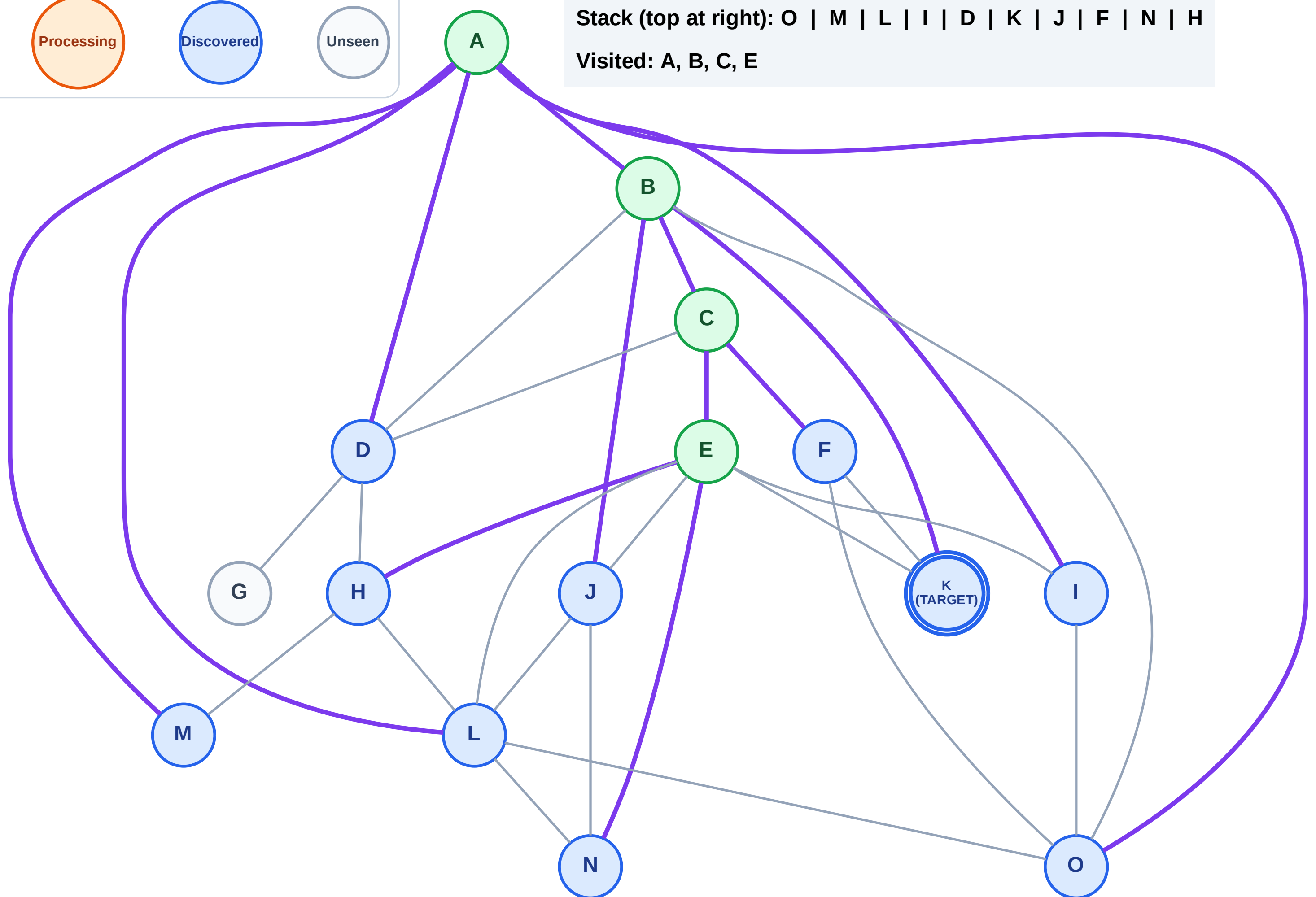
Visited: A, B, C



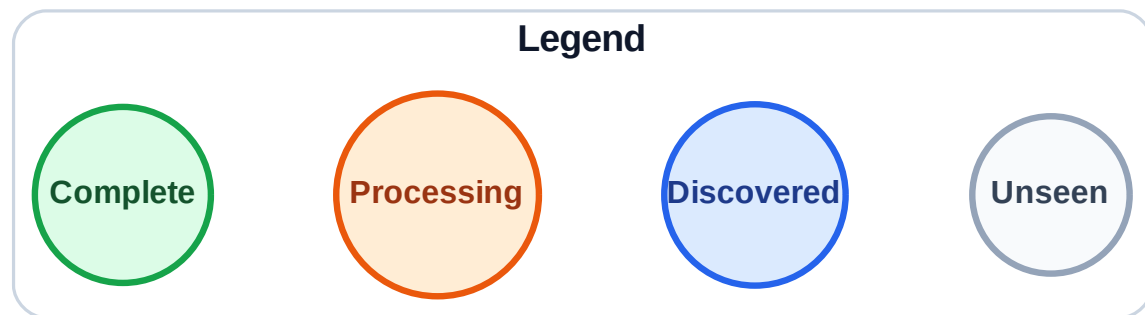
Step 9: Discover N, H from E

Stack (top at right): O | M | L | I | D | K | J | F | N | H

Unseen

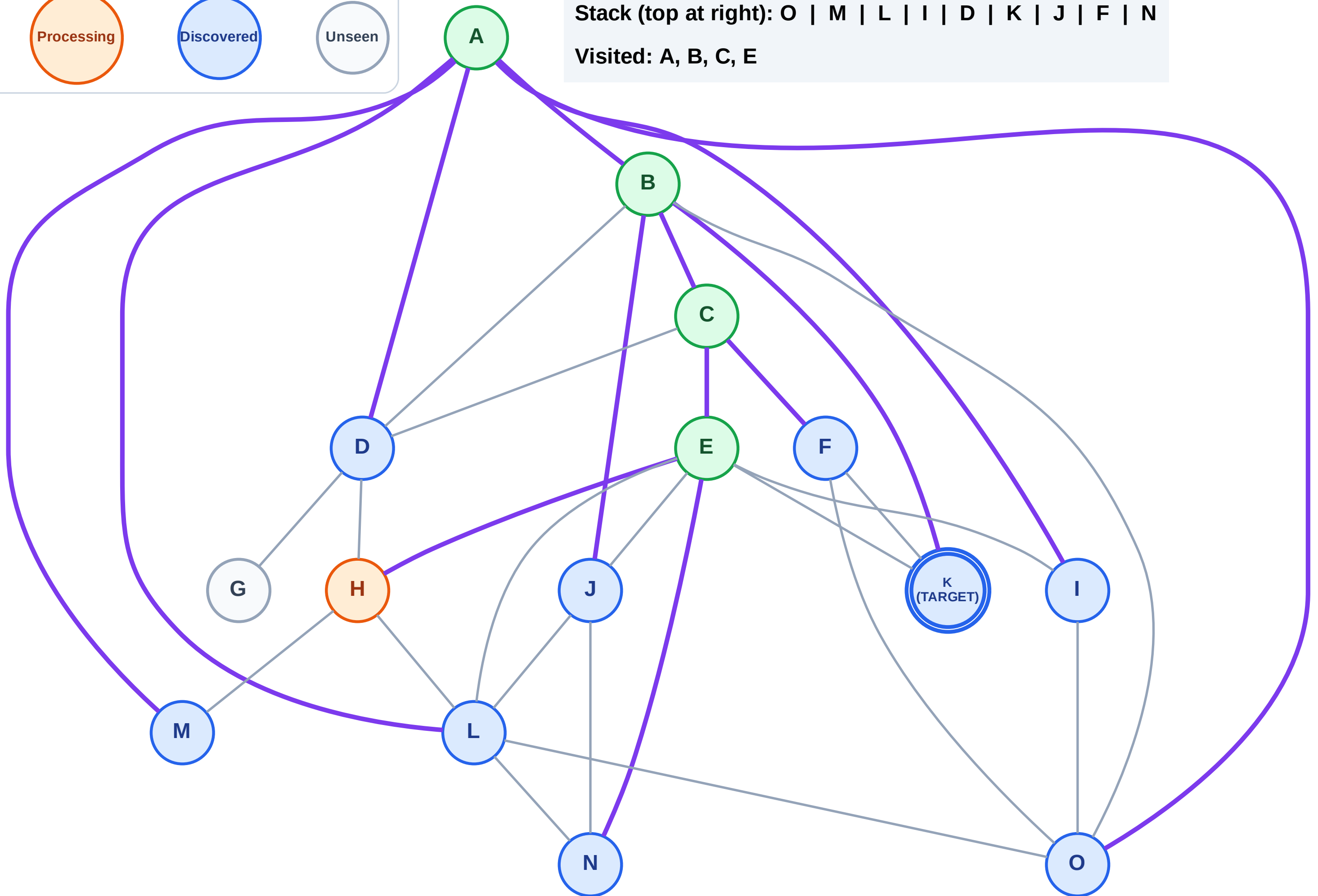


Step 10: Process node H



Stack (top at right): O | M | L | I | D | K | J | F | N

Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from H

Legend

Complete

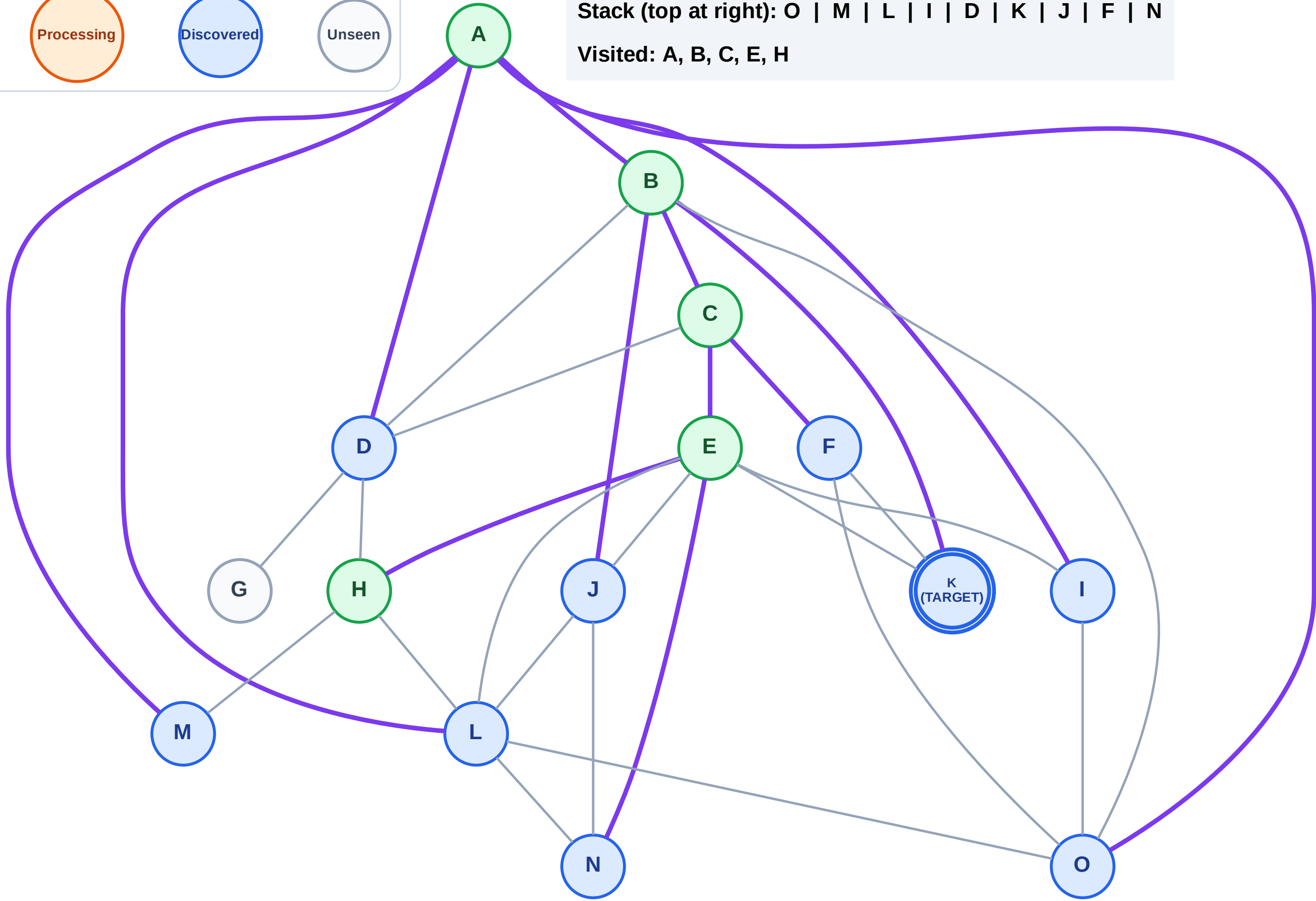
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J | F | N

Visited: A, B, C, E, H



DFS Traversal

Step 12: Process node N

Legend

Complete

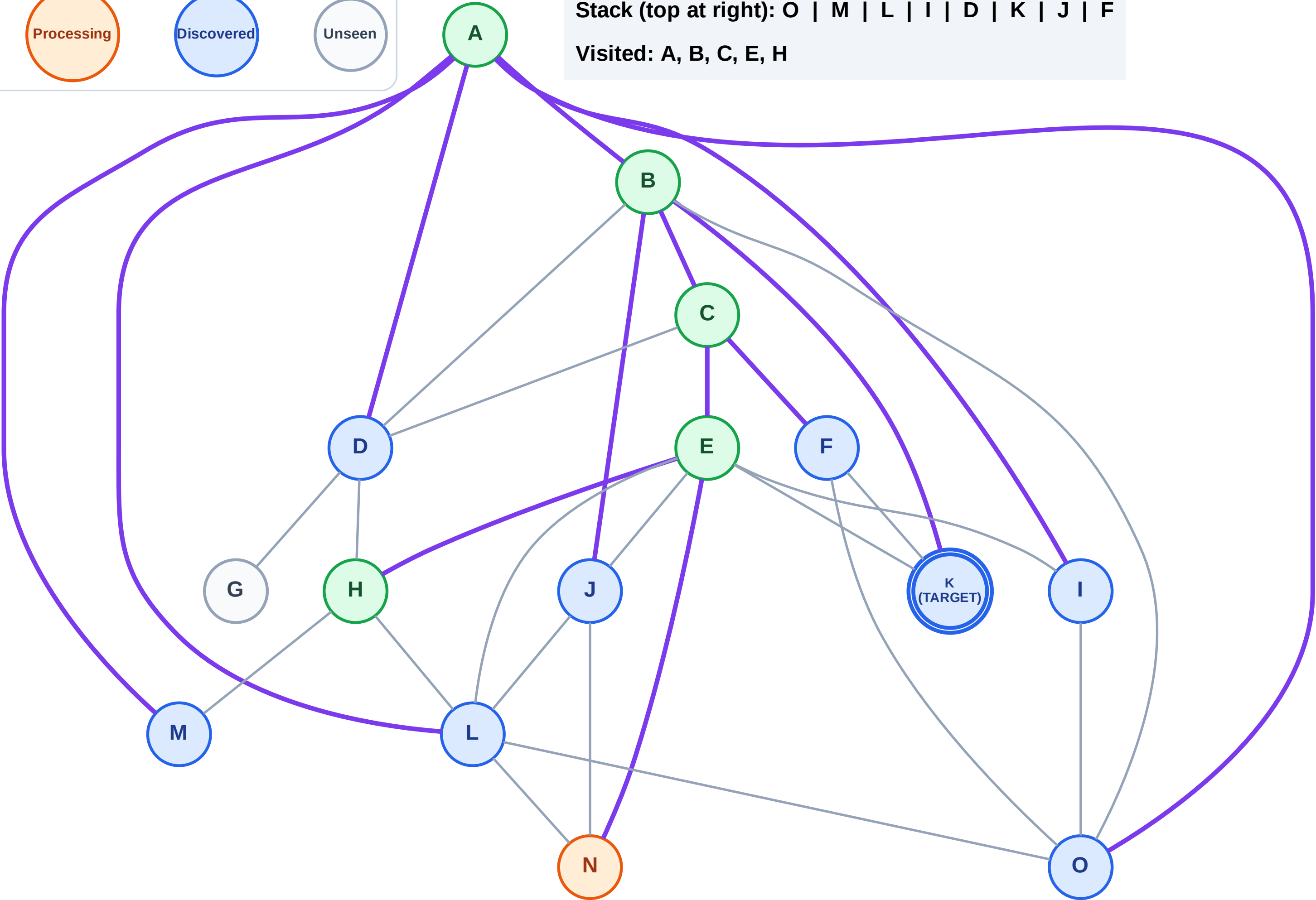
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J | F

Visited: A, B, C, E, H



DFS Traversal

Step 13: No new neighbors from N

Legend

Complete

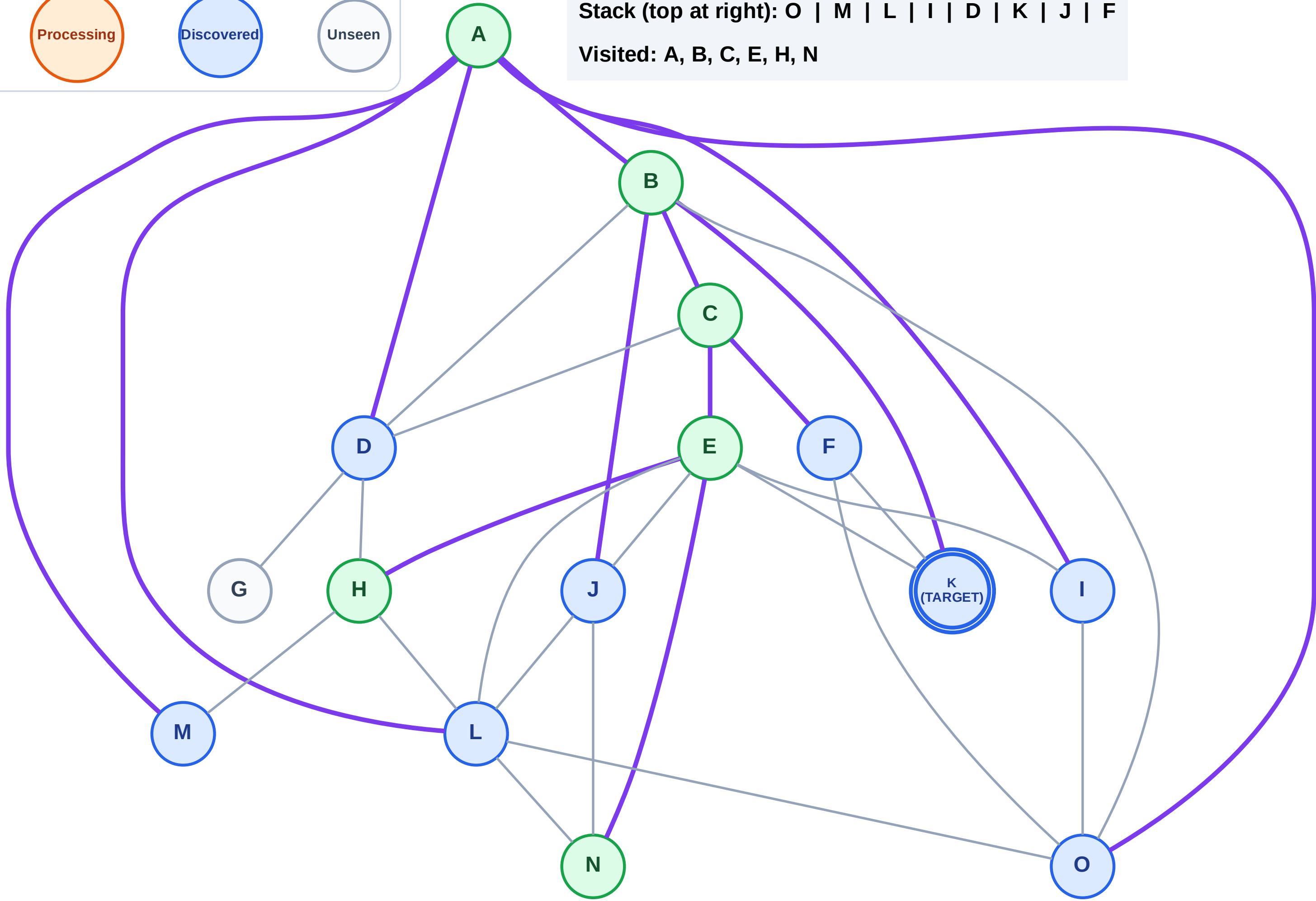
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J | F

Visited: A, B, C, E, H, N



DFS Traversal

Step 14: Process node F

Legend

Complete

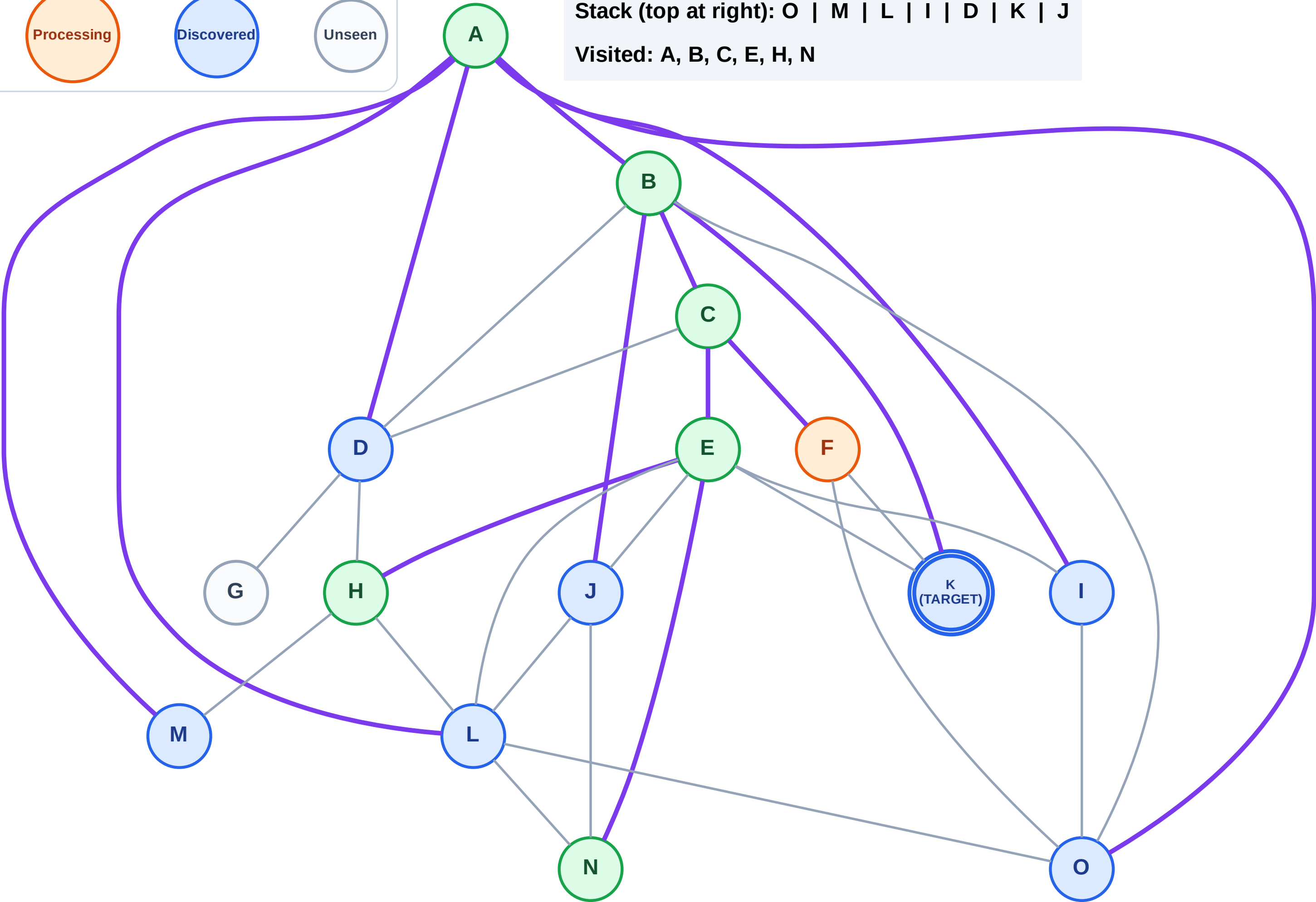
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J

Visited: A, B, C, E, H, N



DFS Traversal

Step 15: No new neighbors from F

Legend

Complete

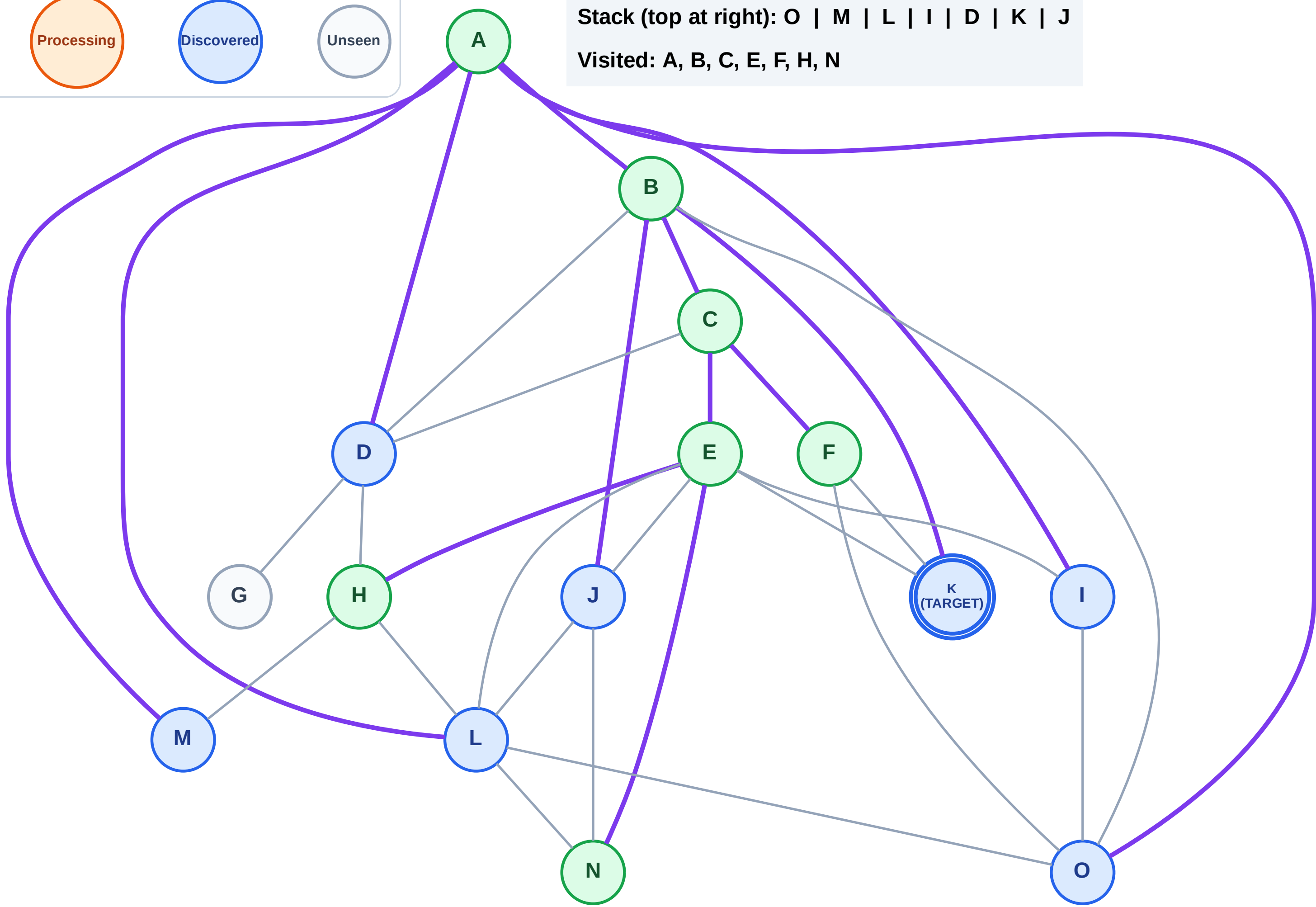
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K | J

Visited: A, B, C, E, F, H, N



DFS Traversal

Step 16: Process node J

Legend

Complete

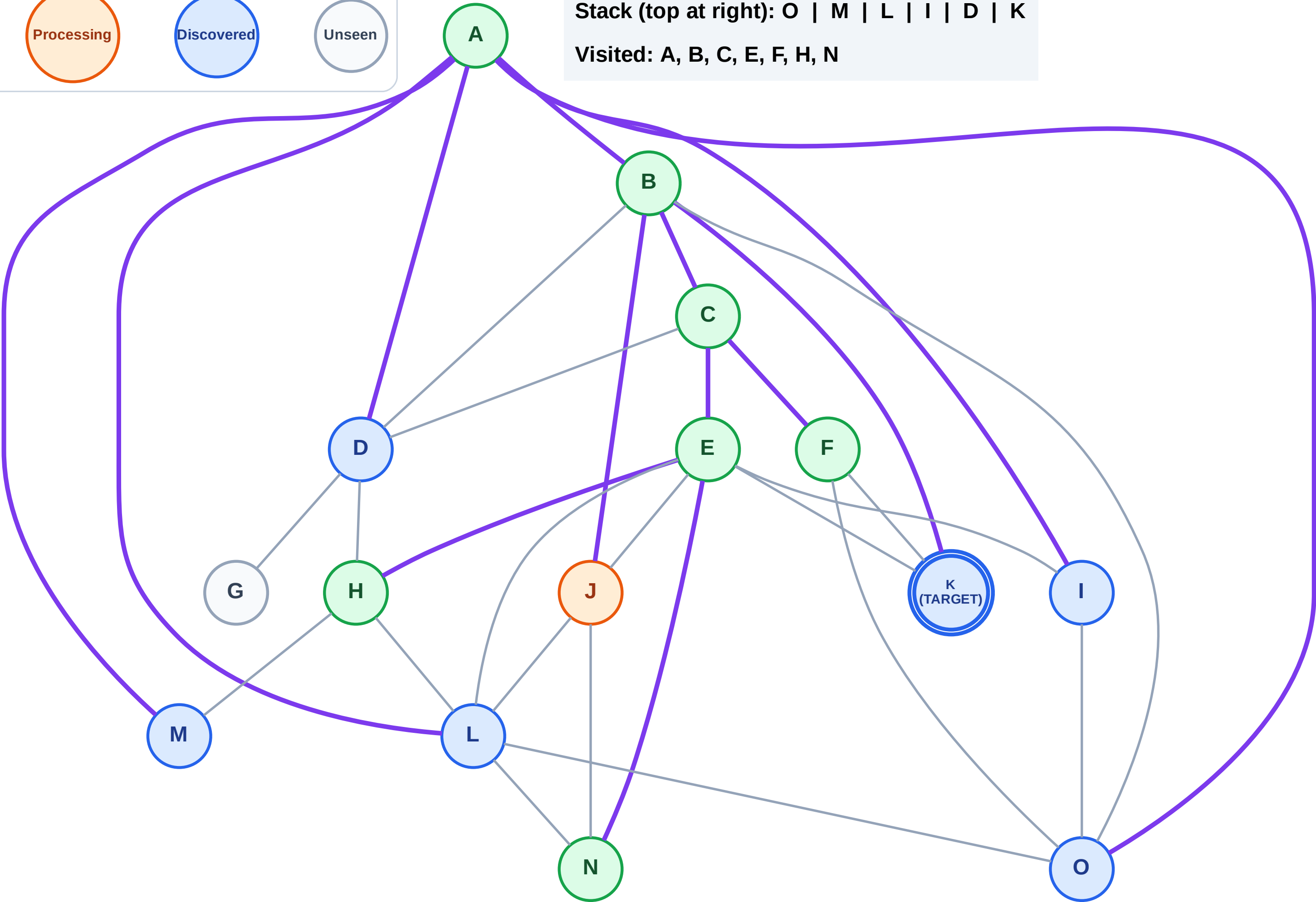
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K

Visited: A, B, C, E, F, H, N



DFS Traversal

Step 17: No new neighbors from J

Legend

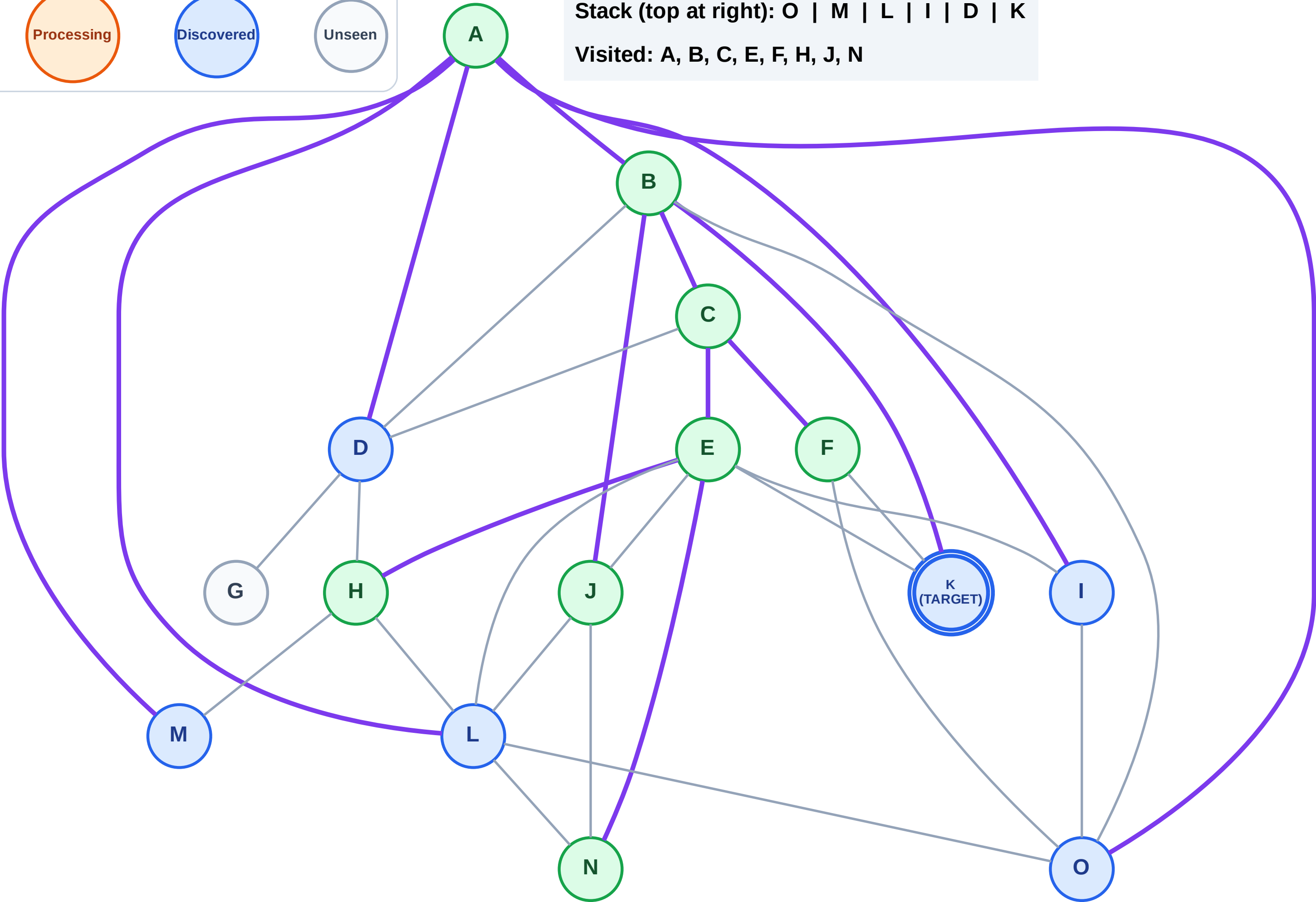
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D | K
Visited: A, B, C, E, F, H, J, N



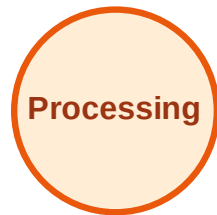
DFS Traversal

Step 18: Process node K

Legend



Complete



Processing



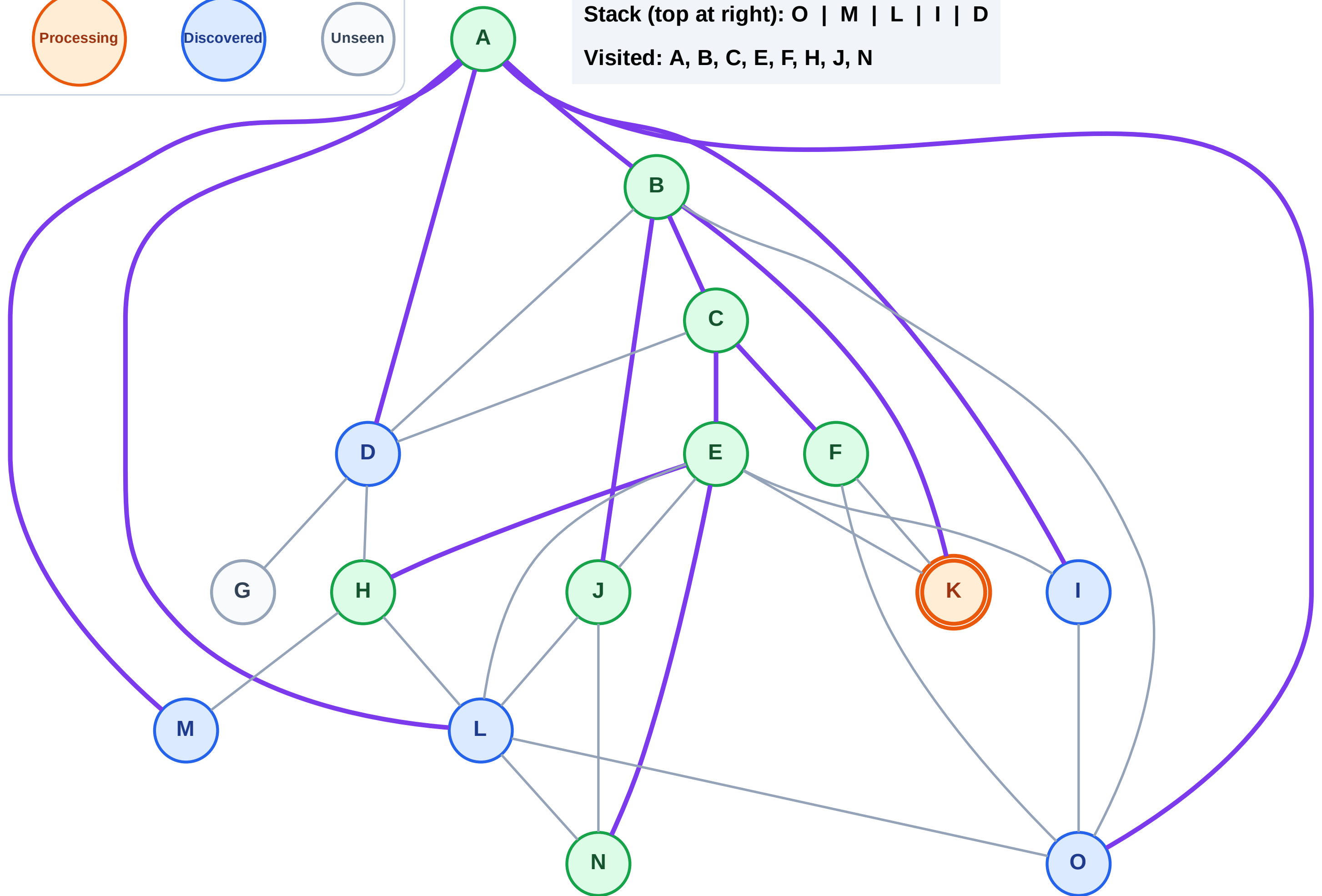
Discovered



Unseen

Stack (top at right): O | M | L | I | D

Visited: A, B, C, E, F, H, J, N



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | D
Visited: A, B, C, E, F, H, J, K, N

